

DpDp

Hei Hao

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1 拉氏量和场定义

正粒子场定义为

$$\begin{aligned} P_a^{*(Q)}(\alpha) &= \sum_{\xi} \int \frac{\sqrt{M_D \cdot d^3 p}}{\sqrt{2E_p}(2\pi)^3} h_{\xi}^a e^{-\alpha p x} \\ P_a^{(Q)} &= \int \frac{\sqrt{M_D d^3 p}}{\sqrt{2E_p}(2\pi)^3} g^a e^{-i p x} \\ P_a^{*(\bar{Q})}(\alpha) &= \sum_{\xi} \int \frac{\sqrt{M_D \cdot d^3 p}}{\sqrt{2E_p}(2\pi)^3} h_{\xi}^{a\dagger} e^{\alpha p x} \\ P_a^{(\bar{Q})} &= \int \frac{\sqrt{M_D d^3 p}}{\sqrt{2E_p}(2\pi)^3} g^{a\dagger} e^{i p x} \end{aligned}$$

拉氏量

$$\begin{aligned} \mathcal{L}_1^+ &= \frac{i v^{\nu}}{2 f_0^2} D^+ D^{+\dagger} (\pi^+ \partial_{\nu} \pi^- - \pi^- \partial_{\nu} \pi^+) \\ \mathcal{L}_2^+ &= -\frac{\sqrt{2} g}{f_0} D^+ D^{*0\dagger\mu} \partial_{\mu} \pi^- \\ \mathcal{L}_3^+ &= -\frac{\sqrt{2} g}{f_0} D^{*0\mu} D^{+\dagger} \partial_{\mu} \pi^+ \end{aligned}$$

π 场

$$\begin{aligned} \pi^+ &= \int d^3 p \frac{1}{\sqrt{2E_p}(2\pi)^3} (a_p e^{i p x} + b_p^{\dagger} e^{-i p x}) \\ \pi^- &= \int d^3 p \frac{1}{\sqrt{2E_p}(2\pi)^3} (a_p^{\dagger} e^{-i p x} + b_p e^{i p x}) \end{aligned}$$

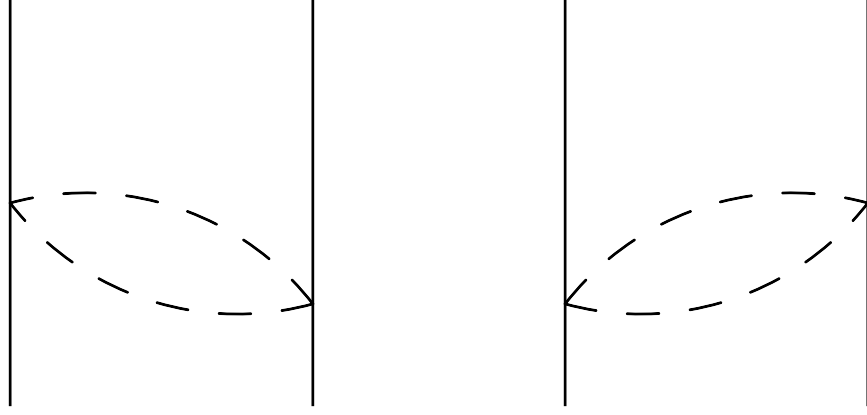


图 1: football

2 football 图

$$\begin{aligned}
W|D^+D^+\rangle &= -\frac{iv^\nu}{2f_0^2}d^3x \int \frac{d^3p_3\sqrt{M_D}}{\sqrt{2E_3}(2\pi)^3}g_3e^{-ip_3x} \int \frac{d^3p_6\sqrt{M_D}}{\sqrt{2E_6}(2\pi)^3}g_6^\dagger e^{ip_6x} \\
&\left(\int d^3p_4\frac{1}{\sqrt{2E_4}(2\pi)^3}(a_4e^{-ip_4x}+b_4^\dagger e^{ip_4x}) \int d^3p_5\frac{1}{\sqrt{2E_5}(2\pi)^3}(ip_5^\nu a_5^\dagger e^{ip_5x}-ip_5^\nu b_5e^{-ip_5x})\right. \\
&\left.-\int d^3p_5\frac{1}{\sqrt{2E_5}(2\pi)^3}(a_5^\dagger e^{ip_5x}+b_5e^{-ip_5x}) \int d^3p_4\frac{1}{\sqrt{2E_4}(2\pi)^3}(-ip_4^\nu a_4e^{-ip_4x}+ip_4^\nu b_4^\dagger e^{ip_4x})\right) \\
&= \frac{v^\nu}{2f_0^2}(p_5^\nu-p_4^\nu) \int d^3x d^3p_3 d^3p_6 d^3p_4 d^3p_5 \frac{m_D}{\sqrt{(2\pi)^{12}2E_32E_42E_52E_6}} g_3g_6^\dagger a_5^\dagger b_4^\dagger e^{i(p_6-p_3+p_4+p_5)} g_1^\dagger g_2^\dagger |0\rangle \\
&= \frac{v^\nu}{2f_0^2}(p_5^\nu-p_4^\nu) \int d^3p_6 d^3p_4 d^3p_5 \frac{m_D}{\sqrt{(2\pi)^62E_32E_42E_52E_6}} \\
&(g_6^\dagger g_2^\dagger a_5^\dagger b_4^\dagger \delta^{(3)}(6-1+4+5) + g_6^\dagger g_1^\dagger a_5^\dagger b_4^\dagger \delta^{(3)}(6-2+4+5))|0\rangle
\end{aligned}$$

$$\begin{aligned}
GW|D^+D^+\rangle &= \frac{v^\nu}{2f_0^2}(p_5^\nu-p_4^\nu) \int d^3p_6 d^3p_4 d^3p_5 \left(\frac{m_D}{\sqrt{(2\pi)^62E_12E_42E_52E_6}} \frac{g_6^\dagger g_2^\dagger a_5^\dagger b_4^\dagger \delta^{(3)}(6-1+4+5)}{z-E_6-E_2-E_4-E_5}\right. \\
&\left. + \frac{m_D}{\sqrt{(2\pi)^62E_22E_42E_52E_6}} \frac{g_6^\dagger g_1^\dagger a_5^\dagger b_4^\dagger \delta^{(3)}(6-2+4+5)}{z-E_6-E_1-E_4-E_5}\right)|0\rangle
\end{aligned}$$

$$\begin{aligned}
\langle D^+D^+|W &= \langle 0|\frac{v^\nu}{2f_0^2}(p_9^\nu-p_8^\nu) \int d^3p_7 d^3p_8 d^3p_9 \frac{m_D}{\sqrt{(2\pi)^62E_72E_82E_92E_{10}}} \\
&(g_7g_{11}a_9b_8\delta^{(3)}(12-7-9-8) + g_7g_{12}a_9b_8\delta^{(3)}(11-7-9-8))
\end{aligned}$$

$$\begin{aligned}
& \langle D^+ D^+ | W G W | D^+ D^+ \rangle \\
&= \frac{p_5 \cdot v - p_4 \cdot v}{4f_0^4} \int d^3 p_6 d^3 p_4 d^3 p_5 d^3 p_7 d^3 p_8 d^3 p_9 \frac{m_D^2}{\sqrt{(2\pi)^{12} 2E_7 2E_8 2E_9 2E_{10} 2E_3 2E_4 2E_5 2E_6}} \\
& \left(\frac{g_7 g_{11} a_9 b_8 \delta^{(3)}(12-7-9-8) g_6^\dagger g_2^\dagger a_5^\dagger b_4^\dagger \delta^{(3)}(6-1+4+5)}{z - E_6 - E_2 - E_4 - E_5} \right. \\
& + \frac{g_7 g_{11} a_9 b_8 \delta^{(3)}(12-7-9-8) g_6^\dagger g_1^\dagger a_5^\dagger b_4^\dagger \delta^{(3)}(6-2+4+5)}{z - E_6 - E_1 - E_4 - E_5} \\
& + \frac{g_7 g_{12} a_9 b_8 \delta^{(3)}(11-7-9-8) g_6^\dagger g_2^\dagger a_5^\dagger b_4^\dagger \delta^{(3)}(6-1+4+5)}{z - E_6 - E_2 - E_4 - E_5} \\
& \left. + \frac{g_7 g_{12} a_9 b_8 \delta^{(3)}(11-7-9-8) g_6^\dagger g_1^\dagger a_5^\dagger b_4^\dagger \delta^{(3)}(6-2+4+5)}{z - E_6 - E_1 - E_4 - E_5} \right) \\
&= \frac{(p_5 \cdot v - p_4 \cdot v)^2}{4f_0^4} \int d^3 p_4 \frac{m_D^2}{(2\pi)^6} \frac{1}{2E_4 2E_5 \sqrt{2E_1 2E_2 2E_3 2E_4}} \left(\frac{\delta^{(3)}(12+11-1-2)}{z - E_{11} - E_2 - E_4 - E_5} \right. \\
& + \frac{\delta^{(3)}(12+11-1-2)}{z - E_{11} - E_1 - E_4 - E_5} + \frac{\delta^{(3)}(12+11-1-2)}{z - E_{12} - E_2 - E_4 - E_5} + \frac{\delta^{(3)}(12+11-1-2)}{z - E_{12} - E_1 - E_4 - E_5} \left. \right)
\end{aligned}$$

3 Triangle 图

$$\begin{aligned}
W | D^+ D^+ \rangle &= \frac{i\sqrt{2}g}{f_0} \sum \int d^3 x d^3 p_3 d^3 p_4 d^3 p_5 \frac{\sqrt{m_D} m_{D^*}}{\sqrt{(2\pi)^9 2E_3 2E_4 2E_5}} p_4^\mu \epsilon_5^{*\mu} \\
& g_3 h_5^\dagger (a_4^\dagger e^{i(p_5 - p_3 + p_4)x} - b_4 e^{i(p_5 - p_3 - p_4)x}) g_1^\dagger g_2^\dagger | 0 \rangle \\
&= \frac{i\sqrt{2}g}{f_0} \sum \int d^3 p_4 d^3 p_5 \frac{\sqrt{m_D} m_{D^*}}{\sqrt{(2\pi)^3 2E_3 2E_4 2E_5}} p_4^\mu \epsilon_5^{*\mu} \\
& (h_5^\dagger g_1^\dagger a_4^\dagger \delta^{(3)}(5-2+4) + h_5^\dagger g_2^\dagger a_4^\dagger \delta^{(3)}(5-1+4)) | 0 \rangle
\end{aligned}$$

$$\begin{aligned}
G W | D^+ D^+ \rangle &= \frac{i\sqrt{2}g}{f_0} \sum \int d^3 p_4 d^3 p_5 \frac{\sqrt{m_D} m_{D^*}}{\sqrt{(2\pi)^3 2E_3 2E_4 2E_5}} p_4^\mu \epsilon_5^{*\mu} \\
& \left(\frac{h_5^\dagger g_1^\dagger a_4^\dagger}{z - E_5 - E_1 - E_4} \delta^{(3)}(5-2+4) + \frac{h_5^\dagger g_2^\dagger a_4^\dagger}{z - E_5 - E_2 - E_4} \delta^{(3)}(5-1+4) \right) | 0 \rangle
\end{aligned}$$

$$\begin{aligned}
WGW|D^+D^+\rangle &= -\frac{iv^\nu}{2f_0^2} \int d^3x d^3p_6 d^3p_7 d^3p_8 d^3p_9 \frac{m_D^2}{\sqrt{(2\pi)^{12} 2E_6 2E_7 2E_8 2E_9}} \\
&g_6 e^{-ip_6 x} g_9^\dagger e^{ip_9 x} [(a_7 e^{-ip_7 x} + b_7^\dagger e^{ip_7 x})(ip_8^\nu a_8^\dagger e^{ip_8 x} \\
&- ip_8^\nu b_8 e^{-ip_8 x}) - (a_8^\dagger e^{ip_8 x} + b_8 e^{-ip_8 x})(-ip_7^\nu a_7 e^{-ip_7 x} + ip_7^\nu b_7^\dagger e^{ip_7 x})] \\
&\frac{i\sqrt{2}g}{f_0} \sum \int d^3p_4 d^3p_5 \frac{\sqrt{m_D} m_{D^*}}{\sqrt{(2\pi)^3 2E_3 2E_4 2E_5}} p_4^\mu \epsilon_5^{*\mu} \\
&(\frac{h_5^\dagger g_1^\dagger a_4^\dagger}{z - E_5 - E_1 - E_4} \delta^{(3)}(5 - 2 + 4) + \frac{h_5^\dagger g_2^\dagger a_4^\dagger}{z - E_5 - E_2 - E_4} \delta^{(3)}(5 - 1 + 4)) |0\rangle \\
&= \frac{v^\nu \cdot (p_8^\nu + p_9^\nu)}{2f_0^2} \frac{i\sqrt{2}g}{f_0} \sum \int d^3p_8 d^3p_9 d^3p_4 d^3p_5 \frac{m_D^2}{\sqrt{(2\pi)^3 2E_6 2E_7 2E_8 2E_9}} \frac{\sqrt{m_D} m_{D^*}}{\sqrt{(2\pi)^3 2E_3 2E_4 2E_5}} p_4^\mu \epsilon_5^{*\mu} \\
&(\frac{g_9^\dagger h_5^\dagger a_8^\dagger}{z - E_5 - E_1 - E_4} \delta(6 - 1) \delta(7 - 4) \delta(9 - 6 - 7 + 8) \delta(5 - 2 + 4) \\
&+ \frac{g_9^\dagger h_5^\dagger a_8^\dagger}{z - E_5 - E_2 - E_4} \delta(6 - 2) \delta(7 - 4) \delta(9 - 6 - 7 + 8) \delta(5 - 1 + 4)) |0\rangle \\
\\
GWGW|D^+D^+\rangle &= \frac{iv^\nu}{2f_0^2} \frac{i\sqrt{2}g}{f_0} \sum \int d^3p_8 d^3p_9 d^3p_4 d^3p_5 \frac{m_D^2}{\sqrt{(2\pi)^9 2E_6 2E_7 2E_8 2E_9}} \frac{\sqrt{m_D} m_{D^*}}{\sqrt{(2\pi)^3 2E_3 2E_4 2E_5}} p_4^\mu \epsilon_5^{*\mu} \\
&(\frac{g_9^\dagger h_5^\dagger a_8^\dagger}{z - E_5 - E_1 - E_4} \delta(6 - 1) \delta(7 - 4) \delta(9 - 6 - 7 + 8) \delta(5 - 2 + 4) \\
&+ \frac{g_9^\dagger h_5^\dagger a_8^\dagger}{z - E_5 - E_2 - E_4} \delta(6 - 2) \delta(7 - 4) \delta(9 - 6 - 7 + 8) \delta(5 - 1 + 4)) \frac{1}{z - E_5 - E_9 - E_8} |0\rangle \\
\\
\langle D^+D^+|W &= \langle 0| \frac{i\sqrt{2}g}{f_0} \sum \int d^3p_{10} d^3p_{11} \frac{\sqrt{m_D} m_{D^*}}{\sqrt{(2\pi)^3 2E_3 2E_4 2E_5}} p_{11}^\mu \epsilon_{10}^\mu \\
&(h_{10} g_{13} a_{11} \delta^{(3)}(14 - 10 - 11) + h_{10} g_{14} a_{11} \delta^{(3)}(13 - 10 - 11)) \\
\\
\langle D^+D^+|WGWGW|D^+D^+\rangle &= \frac{g^2}{f_0^4} v^\nu \cdot (p_8^\nu + p_7^\nu) p_{11}^\mu \epsilon_{10}^\mu p_4^\mu \epsilon_5^{*\mu} \sum \int d^3p_{10} d^3p_{11} d^3p_8 d^3p_7 d^3p_4 d^3p_5 \\
&\frac{m_D^2 m_{D^*}}{\sqrt{(2\pi)^{12} 2E_{10} 2E_{11} 2E_{12} 2E_6 2E_7 2E_8 2E_9 2E_3 2E_4 2E_5}} [\\
&\frac{h_{10} g_{13} a_{11} g_9^\dagger h_5^\dagger a_8^\dagger}{(z - E_5 - E_1 - E_4)(z - E_5 - E_9 - E_8)} \delta^{(3)}(14 - 10 - 11) \delta(6 - 1) \delta(7 - 4) \delta(9 - 6 - 7 + 8) \delta(5 - 2 + 4) \\
&+ \frac{h_{10} g_{13} a_{11} g_9^\dagger h_5^\dagger a_8^\dagger}{(z - E_5 - E_2 - E_4)(z - E_5 - E_9 - E_8)} \delta^{(3)}(14 - 10 - 11) \delta(6 - 2) \delta(7 - 4) \delta(9 - 6 - 7 + 8) \delta(5 - 1 + 4) \\
&+ \frac{h_{10} g_{14} a_{11} g_9^\dagger h_5^\dagger a_8^\dagger}{(z - E_5 - E_1 - E_4)(z - E_5 - E_9 - E_8)} \delta^{(3)}(13 - 10 - 11) \delta(6 - 1) \delta(7 - 4) \delta(9 - 6 - 7 + 8) \delta(5 - 2 + 4) \\
&+ \frac{h_{10} g_{14} a_{11} g_9^\dagger h_5^\dagger a_8^\dagger}{(z - E_5 - E_2 - E_4)(z - E_5 - E_9 - E_8)} \delta^{(3)}(13 - 10 - 11) \delta(6 - 2) \delta(7 - 4) \delta(9 - 6 - 7 + 8) \delta(5 - 1 + 4)] \\
&= -\frac{g^2}{f_0^4} p_{11}^\mu \epsilon_{10}^\mu p_4^\mu \epsilon_5^{*\mu} \sum \int d^3p_4 (\frac{(v \cdot p_8 + v \cdot p_7) m_D^2 m_{D^*}}{32E_8 E_5 E_4 \sqrt{(2\pi)^{12} E_{14} E_{13} E_1 E_2}} \frac{4}{(z - E_5 - E_1 - E_4)(z - E_5 - E_{13} - E_8)}
\end{aligned}$$

$$GW|D^+D^+\rangle = \frac{i\sqrt{2}g}{f_0} \sum \int d^3p_4 d^3p_5 \frac{\sqrt{m_D}m_{D^*}}{\sqrt{(2\pi)^3 2E_3 2E_4 2E_5}} p_4^\mu \epsilon_5^{*\mu} \\ \left(\frac{h_5^\dagger g_1^\dagger a_4^\dagger}{z - E_5 - E_1 - E_4} \delta^{(3)}(5 - 2 + 4) + \frac{h_5^\dagger g_2^\dagger a_4^\dagger}{z - E_5 - E_2 - E_4} \delta^{(3)}(5 - 1 + 4) \right) |0\rangle$$

$$WGW|D^+D^+\rangle = \frac{i\sqrt{2}g}{f_0} \sum \int d^3x d^3p_6 d^3p_7 d^3p_8 \frac{\sqrt{m_D}m_{D^*}}{\sqrt{(2\pi)^9 2E_6 2E_7 2E_8}} p_7^\mu \epsilon_6^\mu \\ g_8^\dagger h_6(-a_4 e^{i(p_8 - p_6 - p_7)x} + b_7^\dagger e^{i(p_8 - p_6 + p_7)x}) \frac{i\sqrt{2}g}{f_0} \sum \int d^3p_4 d^3p_5 \frac{\sqrt{m_D}m_{D^*}}{\sqrt{(2\pi)^3 2E_3 2E_4 2E_5}} p_4^\mu \epsilon_5^{*\mu} \\ \left(\frac{h_5^\dagger g_1^\dagger a_4^\dagger}{z - E_5 - E_1 - E_4} \delta^{(3)}(5 - 2 + 4) + \frac{h_5^\dagger g_2^\dagger a_4^\dagger}{z - E_5 - E_2 - E_4} \delta^{(3)}(5 - 1 + 4) \right) |0\rangle \\ = -\frac{2g^2}{f_0^2} \sum \int d^3p_7 d^3p_8 d^3p_4 d^3p_5 \frac{m_D m_{D^*} p_7^\mu \epsilon_6^\mu p_4^\mu \epsilon_5^{*\mu}}{\sqrt{(2\pi)^6 2E_6 2E_7 2E_8 2E_3 2E_4 2E_5}} \\ \left[\frac{g_8^\dagger g_1^\dagger a_4^\dagger b_7^\dagger}{z - E_5 - E_1 - E_4} \delta^{(3)}(5 - 2 + 4) \delta(6 - 5) + \frac{g_8^\dagger g_2^\dagger a_4^\dagger b_7^\dagger}{z - E_5 - E_2 - E_4} \delta^{(3)}(5 - 1 + 4) \delta(6 - 5) \right] |0\rangle$$

$$GWGW|D^+D^+\rangle = -\frac{2g^2}{f_0^2} \sum \int d^3p_7 d^3p_8 d^3p_4 d^3p_5 \frac{m_D m_{D^*} p_7^\mu \epsilon_6^\mu p_4^\mu \epsilon_5^{*\mu}}{\sqrt{(2\pi)^6 2E_6 2E_7 2E_8 2E_3 2E_4 2E_5}} \\ \left[\frac{g_8^\dagger g_1^\dagger a_4^\dagger b_7^\dagger}{(z - E_5 - E_1 - E_4)(z - E_8 - E_1 - E_4 - E_7)} \delta^{(3)}(5 - 2 + 4) \delta(6 - 5) \right. \\ \left. + \frac{g_8^\dagger g_2^\dagger a_4^\dagger b_7^\dagger}{(z - E_5 - E_2 - E_4)(z - E_8 - E_2 - E_4 - E_7)} \delta^{(3)}(5 - 1 + 4) \delta(6 - 5) \right] \delta^{(3)}(8 - 6 + 7) |0\rangle$$

$$W|D^+D^+\rangle = \frac{v^\nu}{2f_0^2} (p_{11}^\nu - p_{10}^\nu) \int d^3p_9 d^3p_{10} d^3p_{11} \frac{m_D}{\sqrt{(2\pi)^6 2E_9 2E_{10} 2E_{11} 2E_{12}}} \\ (g_{13}g_9 a_{10} b_{11} \delta^{(3)}(14 - 9 - 10 - 11) + g_{14}g_9 a_{10} b_{11} \delta^{(3)}(13 - 9 - 10 - 11)) \delta^{(3)}(8 - 6 + 7) |0\rangle$$

$$\langle D^+D^+ | WGWGW | D^+D^+ \rangle = \frac{g^2 v^\nu}{f_0^4} (p_{11}^\nu - p_{10}^\nu) \sum \int d^3p_4 \frac{m_D^2 m_{D^*}^2 p_{11}^\mu \epsilon_{12}^\mu p_8^\mu \epsilon_9^{*\mu}}{32E_6 E_4 E_7 \sqrt{(2\pi)^6 E_1 E_2 E_{13} E_{14}}} \\ \left[\frac{1}{(z - E_5 - E_1 - E_4)(z - E_{13} - E_1 - E_4 - E_7)} + \frac{1}{(z - E_5 - E_1 - E_4)(z - E_{14} - E_1 - E_4 - E_7)} \right. \\ \left. + \frac{1}{(z - E_5 - E_2 - E_4)(z - E_{13} - E_2 - E_4 - E_7)} + \frac{1}{(z - E_5 - E_2 - E_4)(z - E_{14} - E_2 - E_4 - E_7)} \right]$$

同理可得

$$\langle D^+D^+ | WGWGW | D^+D^+ \rangle = \frac{g^2 v^\nu}{f_0^4} (p_4^\nu - p_5^\nu) \sum \int d^3p_4 \frac{m_D^2 m_{D^*}^2 p_{11}^\mu \epsilon_{12}^\mu p_8^\mu \epsilon_9^{*\mu}}{32E_9 E_8 E_{11} \sqrt{(2\pi)^6 E_1 E_2 E_{13} E_{14}}} \\ \left[\frac{1}{(z - E_6 - E_1 - E_4 - E_5)(z - E_9 - E_{13} - E_5)} + \frac{1}{(z - E_6 - E_2 - E_4 - E_5)(z - E_9 - E_{13} - E_5)} \right. \\ \left. + \frac{1}{(z - E_6 - E_1 - E_4 - E_5)(z - E_9 - E_{14} - E_5)} + \frac{1}{(z - E_6 - E_2 - E_4 - E_5)(z - E_9 - E_{14} - E_5)} \right]$$

4 Box 图

$$\begin{aligned}
W|D^+D^+\rangle &= \frac{i\sqrt{2}g}{f_0} \sum \int d^3x d^3p_3 d^3p_4 d^3p_5 \frac{\sqrt{m_D m_{D^*}}}{\sqrt{(2\pi)^9 2E_3 2E_4 2E_5}} p_4^\mu \epsilon_5^{*\mu} \\
&\quad g_3 h_5^\dagger (a_4^\dagger e^{i(p_5 - p_3 + p_4)x} - b_4^\dagger e^{i(p_5 - p_3 - p_4)x}) g_1^\dagger g_2^\dagger |0\rangle \\
&= \frac{i\sqrt{2}g}{f_0} \sum \int d^3p_4 d^3p_5 \frac{\sqrt{m_D m_{D^*}}}{\sqrt{(2\pi)^3 2E_3 2E_4 2E_5}} p_4^\mu \epsilon_5^{*\mu} \\
&\quad (h_5^\dagger g_1^\dagger a_4^\dagger \delta^{(3)}(5-2+4) + h_5^\dagger g_2^\dagger a_4^\dagger \delta^{(3)}(5-1+4)) |0\rangle
\end{aligned}$$

$$\begin{aligned}
GW|D^+D^+\rangle &= \frac{i\sqrt{2}g}{f_0} \sum \int d^3p_4 d^3p_5 \frac{\sqrt{m_D m_{D^*}}}{\sqrt{(2\pi)^3 2E_3 2E_4 2E_5}} p_4^\mu \epsilon_5^{*\mu} \\
&\quad \left(\frac{h_5^\dagger g_1^\dagger a_4^\dagger}{z - E_5 - E_1 - E_4} \delta^{(3)}(5-2+4) + \frac{h_5^\dagger g_2^\dagger a_4^\dagger}{z - E_5 - E_2 - E_4} \delta^{(3)}(5-1+4) \right) |0\rangle
\end{aligned}$$

$$\begin{aligned}
WGW|D^+D^+\rangle &= \frac{i\sqrt{2}g}{f_0} \sum \int d^3x d^3p_6 d^3p_7 d^3p_8 \frac{\sqrt{m_D m_{D^*}}}{\sqrt{(2\pi)^9 2E_6 2E_7 2E_8}} p_7^\mu \epsilon_6^\mu \\
&\quad g_8^\dagger h_6 (-a_4 e^{i(p_8 - p_6 - p_7)x} + b_7^\dagger e^{i(p_8 - p_6 + p_7)x}) \frac{i\sqrt{2}g}{f_0} \sum \int d^3p_4 d^3p_5 \frac{\sqrt{m_D m_{D^*}}}{\sqrt{(2\pi)^3 2E_3 2E_4 2E_5}} p_4^\mu \epsilon_5^{*\mu} \\
&\quad \left(\frac{h_5^\dagger g_1^\dagger a_4^\dagger}{z - E_5 - E_1 - E_4} \delta^{(3)}(5-2+4) + \frac{h_5^\dagger g_2^\dagger a_4^\dagger}{z - E_5 - E_2 - E_4} \delta^{(3)}(5-1+4) \right) |0\rangle \\
&= -\frac{2g^2}{f_0^2} \sum \int d^3p_7 d^3p_8 d^3p_4 d^3p_5 \frac{m_D m_{D^*} p_7^\mu \epsilon_6^\mu p_4^\mu \epsilon_5^{*\mu}}{\sqrt{(2\pi)^6 2E_6 2E_7 2E_8 2E_3 2E_4 2E_5}} \\
&\quad \left[\frac{g_8^\dagger g_1^\dagger a_4^\dagger b_7^\dagger}{z - E_5 - E_1 - E_4} \delta^{(3)}(5-2+4) \delta(6-5) + \frac{g_8^\dagger g_2^\dagger a_4^\dagger b_7^\dagger}{z - E_5 - E_2 - E_4} \delta^{(3)}(5-1+4) \delta(6-5) \right] |0\rangle
\end{aligned}$$

$$\begin{aligned}
GWGW|D^+D^+\rangle &= -\frac{2g^2}{f_0^2} \sum \int d^3p_7 d^3p_8 d^3p_4 d^3p_5 \frac{m_D m_{D^*} p_7^\mu \epsilon_6^\mu p_4^\mu \epsilon_5^{*\mu}}{\sqrt{(2\pi)^6 2E_6 2E_7 2E_8 2E_3 2E_4 2E_5}} \\
&\quad \left[\frac{g_8^\dagger g_1^\dagger a_4^\dagger b_7^\dagger}{(z - E_5 - E_1 - E_4)(z - E_8 - E_1 - E_4 - E_7)} \delta^{(3)}(5-2+4) \delta(6-5) \right. \\
&\quad \left. + \frac{g_8^\dagger g_2^\dagger a_4^\dagger b_7^\dagger}{(z - E_5 - E_2 - E_4)(z - E_8 - E_2 - E_4 - E_7)} \delta^{(3)}(5-1+4) \delta(6-5) \right] \delta^{(3)}(8-6+7) |0\rangle
\end{aligned}$$

$$\begin{aligned}
GWGWGW|D^+D^+\rangle &= \frac{i2\sqrt{2}g^3}{f_0^3} \sum \int d^3p_{11} d^3p_7 d^3p_8 d^3p_4 d^3p_5 \\
&\quad \frac{(m_D m_{D^*})^{3/2} p_8^\mu \epsilon_{11}^\mu p_7^\mu \epsilon_6^\mu p_4^\mu \epsilon_5^{*\mu}}{\sqrt{(2\pi)^9 2E_9 2E_{10} 2E_{11} 2E_6 2E_7 2E_8 2E_3 2E_4 2E_5}} \\
&\quad \left[\frac{h_{11}^\dagger g_8^\dagger a_4^\dagger \delta^{(3)}(5-2+4) \delta(6-5) \delta(9-1) \delta(7-4)}{(z - E_5 - E_1 - E_4)(z - E_8 - E_1 - E_4 - E_7)(z - E_{11} - E_8 - E_4)} \right. \\
&\quad \left. + \frac{h_{11}^\dagger g_8^\dagger a_4^\dagger \delta^{(3)}(5-1+4) \delta(6-5) \delta(9-2) \delta(7-4)}{(z - E_5 - E_2 - E_4)(z - E_8 - E_2 - E_4 - E_7)(z - E_{11} - E_8 - E_4)} \right] \delta^{(3)}(8-6+7) \delta^{(3)}(11-9-10) |0\rangle
\end{aligned}$$

$$\begin{aligned}
\langle D^+ D^+ | W G W G W G W | D^+ D^+ \rangle &= \frac{4g^4}{f_0^4} \sum \int d^3 p_4 \frac{(m_D m_{D^*})^2 p_{13}^\mu \epsilon_{12}^\mu p_8^\mu \epsilon_{11}^\mu p_7^\mu \epsilon_6^\mu p_4^\mu \epsilon_5^{*\mu}}{64(2\pi)^6 E_4 E_7 E_5 E_6 \sqrt{E_1 E_2 E_{15} E_{16}}} \\
&\left[\frac{1}{(z - E_5 - E_1 - E_4)(z - E_{15} - E_1 - E_4 - E_7)(z - E_{11} - E_{15} - E_4)} \right. \\
&+ \frac{1}{(z - E_5 - E_2 - E_4)(z - E_{15} - E_2 - E_4 - E_7)(z - E_{11} - E_{15} - E_4)} \\
&+ \frac{1}{(z - E_5 - E_1 - E_4)(z - E_{16} - E_1 - E_4 - E_7)(z - E_{11} - E_{16} - E_4)} \\
&\left. + \frac{1}{(z - E_5 - E_2 - E_4)(z - E_{16} - E_2 - E_4 - E_7)(z - E_{11} - E_{16} - E_4)} \right] \delta^{(3)}(16 + 15 - 1 - 2)
\end{aligned}$$

5 分波分解

根据 note 中公式 130

$$\begin{aligned}
\langle \lambda'_1 \lambda'_2 | V^J(q', q, E) | \lambda_1 \lambda_2 \rangle &= 2\pi \int_{-1}^1 d(\cos \theta) d_{\lambda'_1 \lambda'}^J(\cos \theta) \langle q' \lambda'_1 \lambda'_2 | V^J(q', q, E) | q \lambda_1 \lambda_2 \rangle \\
\langle L' S J | V^J(q', q, E) | L S J \rangle &= \sum_{\lambda'_1 \lambda'_2 \lambda_1 \lambda_2} \langle L' S J | J M \lambda'_1 \lambda'_2 \rangle \langle \lambda'_1 \lambda'_2 | V^J(q', q, E) | \lambda_1 \lambda_2 \rangle \langle J M \lambda_1 \lambda_2 | L S J \rangle \\
\langle L' S J' | V^J(q', q, E) | L S J \rangle &= \langle L' S J | V^J(q', q, E) | L S J \rangle \delta_{S' S} \delta_{J' J} \\
\langle J M \lambda_1 \lambda_2 | L S J \rangle &= \left(\frac{2L+1}{2J+1} \right)^{1/2} C(L S J; 0, \lambda) C(s_1 s_2 S; \lambda_1, -\lambda_2) \\
&= \left(\frac{2L+1}{2J+1} \right)^{1/2} (L S, 0, \lambda | J \lambda) (s_1 s_2, \lambda_1, -\lambda_2 | S \lambda)
\end{aligned}$$

由于标量介子的自旋以及螺旋度为 0

$$\begin{aligned}
\langle 00 | V^J | 00 \rangle &= 2\pi \int_{-1}^1 d(\cos \theta) d_{00}^J(\theta) \langle p', 00 | V | p, 00 \rangle \\
&= 2\pi \int_{-1}^1 d(\cos \theta) P_J(\cos \theta) \langle p', 00 | V | p, 00 \rangle
\end{aligned} \tag{1}$$

$$\langle L 0 J | V | L 0 J \rangle = \langle J M; L S | J M; 00 \rangle \langle 00 | V^J | 00 \rangle \langle J M; 00 | J M; L S \rangle \tag{2}$$

$$\langle J M; L S | J M; 00 \rangle = \left(\frac{2J+1}{2J+1} \right)^{\frac{1}{2}} \langle J 0; 00 | J 0 \rangle \langle 00; 00 | 00 \rangle = 1$$

$$\langle J M; 00 | J M; J 0 \rangle = (\langle J M; J 0 | J M; 00 \rangle)^* = 1$$

$$\begin{aligned}
\langle J 0 J | V | J 0 J \rangle &= \langle 00 | V^J | 00 \rangle \\
&= 2\pi \int_{-1}^1 dz P_J(z) \langle p', 00 | V | p, 00 \rangle \\
&= 2\pi \int_{-1}^1 dz P_J(z) V_{D^+ D^+}
\end{aligned} \tag{3}$$

其中 $P_J(z)$ 为勒让德函数

6 polarization vector

At the rest frame we have

$$\epsilon_\lambda = (0, \vec{\epsilon}_\lambda)$$

making a lorentz boost to Eq to derive the polarization vector

$$\epsilon_\lambda = \left(\frac{\vec{p} \cdot \vec{\epsilon}_\lambda}{m}, \vec{\epsilon}_\lambda + \frac{\vec{p}(\vec{p} \cdot \vec{\epsilon}_\lambda)}{m(P_0 + m)} \right)$$

where $p = (p_0, \mathbf{P})$ is the particle's 4-momentum and m is the mass of the particle.

自旋为 $J = 1$ 的极化矢量

$$\begin{aligned} \epsilon^\mu(\pm 1) &= \frac{1}{\sqrt{2}}(0, \pm 1, i, 0) \\ \epsilon^\mu(0) &= \frac{1}{\sqrt{2}}(0, 0, 0, -1) \end{aligned}$$

$$\epsilon_{\lambda_1}(p) \cdot \epsilon_{\lambda_3}^\dagger(p'):$$

$$\begin{aligned} \epsilon_{\lambda_1}^\mu(p) \cdot \epsilon_{\lambda_3}^\mu(p') &= \frac{(\vec{p} \cdot \vec{\epsilon}_{\lambda_1})(\vec{p}' \cdot \vec{\epsilon}_{\lambda_3}^\dagger)}{m^2} - (\epsilon_\lambda \cdot \epsilon_{\lambda_3}^\dagger + \frac{(\vec{p}' \cdot \vec{\epsilon}_{\lambda_1})(\vec{p}' \cdot \vec{\epsilon}_{\lambda_3}^\dagger)}{m(P_0 + m)} + \frac{(\vec{p} \cdot \vec{\epsilon}_{\lambda_3}^\dagger)(\vec{p} \cdot \vec{\epsilon}_{\lambda_1})}{m(P_0 + m)}) \\ &\quad + \frac{(\vec{p} \cdot \vec{\epsilon}_{\lambda_1})(\vec{p}' \cdot \vec{\epsilon}_{\lambda_3}^\dagger)}{m^2(P_0 + m)^2} \end{aligned}$$

7 ϵ_{lab}^μ 的推导

7.1 基本定义

静止系 S : (ct, x, y, z)

运动系 S' : (ct', x', y', z') , 以速度 $\vec{v}$ 相对于 S 系运动

沿 x 方向的 Boost 变换:

$$\begin{aligned} ct' &= \gamma(ct - \beta x) \\ x' &= \gamma(x - \beta ct) \\ y' &= y \\ z' &= z \end{aligned}$$

其中 $\beta = v/c$, $\gamma = 1/\sqrt{1 - \beta^2}$ 。

矩阵形式为:

$$\begin{pmatrix} ct' \\ x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix}$$

$$\Lambda = \begin{pmatrix} \gamma & -\gamma\vec{\beta}^T \\ -\gamma\vec{\beta} & I + (\gamma - 1)\frac{\vec{\beta}\vec{\beta}^T}{\beta^2} \end{pmatrix} \quad (4)$$

7.2 实验室系推导

在粒子静止系中

$$\epsilon_{\lambda}^{\prime\mu} = (0, \vec{\epsilon}_{\lambda})$$

粒子在实验室系的四动量记为 $p^{\mu} = (P_0, \vec{p})$, 质量为 m 。定义无量纲速度与洛伦兹因子为

$$\vec{\beta} = \frac{\vec{v}}{c} = \frac{\vec{p}}{P_0}, \quad \gamma = \frac{1}{\sqrt{1 - \beta^2}} = \frac{P_0}{m},$$

(采用自然单位 $c = 1$)。

沿 $\vec{\beta}$ 的洛伦兹 boost 对任意四矢量 $A'^{\mu} = (A'^0, \vec{A}')$ 的变换为

$$\begin{aligned} A^0 &= \gamma(A'^0 + \vec{\beta} \cdot \vec{A}'), \\ \vec{A} &= \vec{A}' + (\gamma - 1) \frac{\vec{\beta}(\vec{\beta} \cdot \vec{A}')}{\beta^2} + \gamma A'^0 \vec{\beta}. \end{aligned}$$

将 $A'^{\mu} = \epsilon_{\lambda}^{\prime\mu} = (0, \vec{\epsilon}_{\lambda})$ 代入, 得到时间分量和空间分量:

时间分量

$$\epsilon_{\lambda, \text{lab}}^0 = \gamma(0 + \vec{\beta} \cdot \vec{\epsilon}_{\lambda}) = \gamma \vec{\beta} \cdot \vec{\epsilon}_{\lambda} = \frac{P_0}{m} \frac{\vec{p} \cdot \vec{\epsilon}_{\lambda}}{P_0} = \frac{\vec{p} \cdot \vec{\epsilon}_{\lambda}}{m}.$$

空间分量

$$\vec{\epsilon}_{\lambda, \text{lab}} = \vec{\epsilon}_{\lambda} + (\gamma - 1) \frac{\vec{\beta}(\vec{\beta} \cdot \vec{\epsilon}_{\lambda})}{\beta^2}.$$

代入 $\vec{\beta} = \frac{\vec{p}}{P_0}$ 和 $\beta^2 = \frac{|\vec{p}|^2}{P_0^2}$, 得

$$\begin{aligned} (\gamma - 1) \frac{\vec{\beta}(\vec{\beta} \cdot \vec{\epsilon}_{\lambda})}{\beta^2} &= \left(\frac{P_0}{m} - 1 \right) \frac{\frac{\vec{p}}{P_0} \cdot \frac{\vec{p} \cdot \vec{\epsilon}_{\lambda}}{P_0}}{|\vec{p}|^2 / P_0^2} \\ &= \frac{P_0 - m}{m} \frac{\vec{p}(\vec{p} \cdot \vec{\epsilon}_{\lambda})}{|\vec{p}|^2}. \end{aligned}$$

利用色散关系 $|\vec{p}|^2 = P_0^2 - m^2 = (P_0 - m)(P_0 + m)$, 有

$$\frac{P_0 - m}{|\vec{p}|^2} = \frac{1}{P_0 + m}.$$

因此

$$(\gamma - 1) \frac{\vec{\beta}(\vec{\beta} \cdot \vec{\epsilon}_{\lambda})}{\beta^2} = \frac{\vec{p}(\vec{p} \cdot \vec{\epsilon}_{\lambda})}{m(P_0 + m)}.$$

于是空间分量为

$$\vec{\epsilon}_{\lambda, \text{lab}} = \vec{\epsilon}_{\lambda} + \frac{\vec{p}(\vec{p} \cdot \vec{\epsilon}_{\lambda})}{m(P_0 + m)}.$$

合并时间分量和空间分量, 得到在实验室系的极化四矢量:

$$\boxed{\epsilon_{\lambda, \text{lab}}^{\mu} = \left(\frac{\vec{p} \cdot \vec{\epsilon}_{\lambda}}{m}, \quad \vec{\epsilon}_{\lambda} + \frac{\vec{p}(\vec{p} \cdot \vec{\epsilon}_{\lambda})}{m(P_0 + m)} \right)}. \quad (14)$$

检验正交性 $p_\mu \epsilon_{\lambda, \text{lab}}^\mu = 0$ 。在静止系 $p'^\mu = (m, \vec{0})$ 且 $p'_\mu \epsilon'^\mu = 0$ ，洛伦兹变换保持内积不变，因此在实验室系仍满足正交条件。

计算其中的动量积分

$$\begin{aligned} I &= \epsilon^i \epsilon^{\dagger j} \int \frac{d^3 l}{(2\pi)^3} \frac{l^i l^j}{\omega_+ \omega_- (\omega_+ + \omega_-)} \\ &= \frac{\epsilon^i \epsilon^{\dagger j}}{\pi} \int_{-\infty}^{\infty} d\lambda \int \frac{d^3 l}{(2\pi)^3} \frac{l^i l^j}{[(l+q)^2 + 4m_1^2 + \lambda^2][(l-q)^2 + 4m_2^2 + \lambda^2]} \end{aligned} \quad (5)$$

$$\begin{aligned} I^0 &= \frac{1}{\pi} \int_{-\infty}^{\infty} d\lambda \int \frac{d^3 l}{(2\pi)^3} \frac{l^i l^j}{[(l+q)^2 + 4m_1^2 + \lambda^2][(l-q)^2 + 4m_2^2 + \lambda^2]} \\ &= \frac{1}{\pi} \int_0^1 dx \int_{-\infty}^{\infty} d\lambda \int \frac{d^3 l}{(2\pi)^3} \frac{l^i l^j}{[l^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + \lambda^2]^2} \\ &= \frac{1}{\pi} \int_0^1 dx \int_{-\infty}^{\infty} d\lambda \int \frac{d^3 l}{(2\pi)^3} \frac{l^i l^j - (2x-1)(l^i q^j + q^i l^j) + (2x-1)^2 q^i q^j}{[l^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + \lambda^2]^2} \end{aligned}$$

分子为 l 的奇数次幂对积分贡献为 0

$$I^0 = \frac{1}{\pi} \int_0^1 dx \int_{-\infty}^{\infty} d\lambda \int \frac{d^3 l}{(2\pi)^3} \frac{l^i l^j + (2x-1)^2 q^i q^j}{[l^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + \lambda^2]^2}$$

$q^i q^j$ 项

$$\begin{aligned} I_q &= -\frac{q^i q^j}{24\pi^2} \left\{ R - \frac{1}{q^4} \left[\frac{5q^4}{3} + 4q^2(m_1^2 + m_2^2) + 4(m_1^2 - m_2^2)^2 \right] + \ln \left(\frac{4m_1 m_2}{\mu^2} \right) \right. \\ &\quad - \frac{m_1^2 - m_2^2}{q^6} [4(m_1^2 - m_2^2)^2 + 6q^2(m_1^2 + m_2^2) + 3q^4] \ln \left(\frac{m_1}{m_2} \right) \\ &\quad \left. + \frac{2L(q)}{q^4} (q^4 + 2q^2(m_1^2 + m_2^2) + 4(m_1^2 - m_2^2)^2) \right\} \end{aligned}$$

分子为 $l^i l^j$ 的项

$$\begin{aligned} I^l &= \frac{1}{\pi} \int_0^1 dx \int_{-\infty}^{\infty} d\lambda \int \frac{d^3 l}{(2\pi)^3} \frac{d^3 l}{[l^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + \lambda^2]^2} dl \\ &= 2 \int_0^1 dx \int \frac{d^4 k}{(2\pi)^4 [k^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2]^2} \end{aligned}$$

$$\begin{aligned}
I^{l,d} &= \mu^{4-d} \int \frac{d^d \tilde{k}}{(2\pi)^d} \frac{\tilde{k}^i \tilde{k}^j}{[\tilde{k}^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2]^2} \\
&= \frac{g^{ij}}{d} \mu^{4-d} \int \frac{d^d \tilde{k}}{(2\pi)^d} \frac{\tilde{k}^i \tilde{k}^j}{[\tilde{k}^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2]^2} \\
&= \frac{g^{ij} \mu^{4-d}}{\Gamma(1 - \frac{d}{2})} \\
&= \frac{2(4\pi)^{\frac{d}{2}} [4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2]^{1-\frac{d}{2}}}{\Gamma(\frac{d}{2} - 1)} \\
&= \frac{g^{ij} \mu^\epsilon}{2(4\pi)^{2-\frac{\epsilon}{2}} [4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2]^{\frac{\epsilon}{2}-1}} \\
&= g^{ij} \frac{4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2}{32\pi^2} (R - 2\ln(\mu) + \ln[4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2])
\end{aligned}$$

所以

$$\begin{aligned}
I^l &= 2g^{ij} \int_0^1 dx \frac{4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2}{32\pi^2} (R - 2\ln(\mu) + \ln[4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2]) \\
&= \frac{g^{ij}}{24\pi^2} \left\{ q^2 + 3(m_1^2 + m_2^2) \left[R + \ln\left(\frac{4m_1 m_2}{\mu^2}\right) \right] \right. \\
&\quad - \frac{m_1^2 - m_2^2}{q^4} ((m_1^2 - m_2^2)^2 + 3q^2(m_1^2 + m_2^2)) \ln\left(\frac{m_1}{m_2}\right) \\
&\quad \left. - \frac{1}{q^2} \left[(m_1^2 - m_2^2)^2 + 4q^2(m_1^2 + m_2^2) + \frac{5}{3}q^4 \right] + 2w^2(q)L(q) \right\}
\end{aligned}$$

最终

$$I = -\epsilon \cdot \epsilon^\dagger * A + (\epsilon \cdot q)(\epsilon^\dagger \cdot q) * B$$

其中

$$\begin{aligned}
A &= \frac{1}{24\pi^2} \left\{ q^2 + 3(m_1^2 + m_2^2) \left[R + \ln\left(\frac{4m_1 m_2}{\mu^2}\right) \right] \right. \\
&\quad - \frac{m_1^2 - m_2^2}{q^4} ((m_1^2 - m_2^2)^2 + 3q^2(m_1^2 + m_2^2)) \ln\left(\frac{m_1}{m_2}\right) \\
&\quad \left. - \frac{1}{q^2} \left[(m_1^2 - m_2^2)^2 + 4q^2(m_1^2 + m_2^2) + \frac{5}{3}q^4 \right] + 2w^2(q)L(q) \right\} \\
B &= -\frac{1}{24\pi^2} \left\{ R - \frac{1}{q^4} \left[\frac{5q^4}{3} + 4q^2(m_1^2 + m_2^2) + 4(m_1^2 - m_2^2)^2 \right] + \ln\left(\frac{4m_1 m_2}{\mu^2}\right) \right. \\
&\quad - \frac{m_1^2 - m_2^2}{q^6} [4(m_1^2 - m_2^2)^2 + 6q^2(m_1^2 + m_2^2) + 3q^4] \ln\left(\frac{m_1}{m_2}\right) \\
&\quad \left. + \frac{2L(q)}{q^4} (q^4 + 2q^2(m_1^2 + m_2^2) + 4(m_1^2 - m_2^2)^2) \right\}
\end{aligned}$$

$$\epsilon^\mu(\pm 1)_i = \frac{1}{\sqrt{2}}(0, \mp 1, -i, 0)$$

$$\epsilon^{\mu\dagger}(\pm 1)_f = \frac{1}{\sqrt{2}}(0, \mp \cos \theta, i, \pm \sin \theta)^T$$

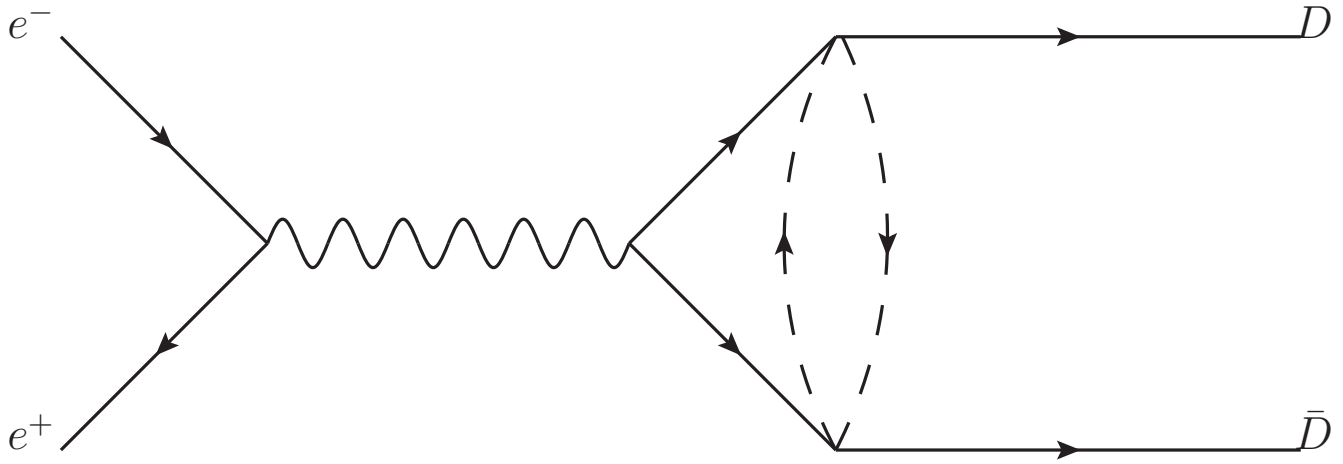


图 2: $e^+e^- \longrightarrow D\bar{D}$

$$\mathcal{L} = -e\bar{\psi}(x)\gamma^\mu\psi(x)A_\mu(x)$$

$D\bar{D}$ 系统的同位旋分解

D 介子构成同位旋二重态:

$$D = \begin{pmatrix} D^+ \\ D^0 \end{pmatrix}, \quad I = \frac{1}{2}$$

其中 $I_3(D^+) = +\frac{1}{2}$, $I_3(D^0) = -\frac{1}{2}$ 。

反 D 介子二重态:

$$\bar{D} = \begin{pmatrix} \bar{D}^0 \\ -D^- \end{pmatrix}, \quad I = \frac{1}{2}$$

其中 $I_3(\bar{D}^0) = +\frac{1}{2}$, $I_3(D^-) = -\frac{1}{2}$ 。

$D\bar{D}$ 系统的同位旋分解为 $\frac{1}{2} \otimes \frac{1}{2} = 1 \oplus 0$:

$$|D\bar{D}, I=1, I_3=+1\rangle = D^+\bar{D}^0 \quad (6)$$

$$|D\bar{D}, I=1, I_3=0\rangle = \frac{1}{\sqrt{2}} (D^0\bar{D}^0 - D^+D^-) \quad (7)$$

$$|D\bar{D}, I=1, I_3=-1\rangle = -D^0D^- \quad (8)$$

$$|D\bar{D}, I=0, I_3=0\rangle = \frac{1}{\sqrt{2}} (D^0\bar{D}^0 + D^+D^-) \quad (9)$$

表 1: $I = 0$ 的同位旋因子表

	中间态	$\pi\pi$	KK	$\eta\eta$	$\pi\eta$
足球图	-	-24	-16	-	-
左三角图	D^*	2	-	-	-
	D_s^*	-	2	-	-
右三角图	D^*	2	-	-	-
	D_s^*	-	2	-	-
平行箱图	D^*D^*	$\frac{9}{16}$	-	$\frac{1}{144}$	$\frac{5}{48}$
	$D_s^*D_s^*$	-	$\frac{1}{2}$	-	-
交叉箱图	D^*D^*	$\frac{1}{8}$	-	$\frac{1}{144}$	-
	$D_s^*D_s^*$	-	-	-	-

表 2: $I = 1$ 的同位旋因子表

	中间态	$\pi\pi$	KK	$\eta\eta$	$\pi\eta$
足球图	-	8	0	-	-
左三角图	D^*	0	-	-	-
	D_s^*	-	0	-	-
右三角图	D^*	0	-	-	-
	D_s^*	-	0	-	-
平行箱图	D^*D^*	$\frac{1}{8}$	-	$\frac{1}{144}$	$-\frac{3}{144}$
	$D_s^*D_s^*$	-	0	-	-
交叉箱图	D^*D^*	$\frac{1}{32}$	-	$\frac{1}{288}$	-
	$D_s^*D_s^*$	-	-	-	-

在矢量连络中引入光子场

$$\begin{aligned} D_\mu &= \partial_\mu + \mathcal{V}_\mu \\ \mathcal{V}_\mu &= \frac{1}{2}(\xi^\dagger(\partial_\mu - ir_\mu)\xi + \xi(\partial_\mu - il_\mu)\xi^\dagger) \\ r_\mu &= l_\mu = -eQA_\mu \end{aligned}$$

$$\begin{aligned} \mathcal{L}_{\gamma D \bar{D}} &= g_{\gamma D \bar{D}} \{D \partial_\mu \bar{D} - \partial_\mu D \bar{D}\} A^\mu, \\ \mathcal{L}_{\gamma D \bar{D}^*} &= -ig_{\gamma D \bar{D}^*} \epsilon_{\alpha\beta\mu\nu} \partial^\alpha A^\beta \partial^\mu \bar{D}^{*\nu} D + \text{h.c.}, \\ \mathcal{L}_{\gamma D^* \bar{D}^*} &= g_{\gamma D^* \bar{D}^*} \{A^\mu (\partial_\mu D^{*\nu} \bar{D}_\nu^* - D^{*\nu} \partial_\mu \bar{D}_\nu^*) \\ &\quad + (\partial_\mu A_\nu D^{*\nu} - A_\nu \partial_\mu D^{*\nu}) \bar{D}^{*\mu} \\ &\quad + D^{*\mu} (A^\nu \partial_\mu \bar{D}_\nu^* - \partial_\mu A_\nu \bar{D}^{*\nu})\}, \end{aligned}$$

考虑单光子贡献, $e^+e^- \rightarrow D\bar{D}$ 过程的振幅可以写为

$$T(0, 0; \lambda_1, \lambda_2) = N_i \frac{\alpha}{2} g(s) \bar{v}^e(-\mathbf{k}, \lambda_2) \gamma_\mu u^e(\mathbf{k}, \lambda_1) \frac{1}{s} (p_3 - p_0)^\mu$$

到 $D^* \bar{D}^*$ 有 $|^1P_1\rangle, |^5P_1\rangle, |^5F_1\rangle$

$$T(\pm 1, \pm 1; \lambda_1, \lambda_2) = N_i \frac{\alpha}{2} g(s) \bar{v}^e(-\mathbf{k}, \lambda_2) \gamma_\mu u^e(\mathbf{k}, \lambda_1) \frac{1}{s} [(p_3 - p_0)^\mu (\epsilon_3^\nu \cdot \epsilon_4^{\dagger\nu}) + \epsilon_3^\nu \cdot p_4^\nu \cdot \epsilon_4^{\dagger\mu} + \epsilon_4^{\dagger\nu} \cdot p_3^\nu \cdot \epsilon_3^\mu]$$

其中 p_3 和 p_4 分别为 $D^+(D^0)$ 和 $D^-(\bar{D}^0)$ 的动量。在质心系中, 定义

$$\begin{aligned} p_3^\mu &= (E, \mathbf{k}') \\ p_4^\mu &= (E, -\mathbf{k}') \end{aligned}$$

$$\begin{aligned} T(0, 0; +, +) &= T(0, 0; -, -) = \frac{\alpha g(s) k' \cos \theta m_e}{s} \\ T(0, 0; +, -) &= -T(0, 0; -, +) = -\frac{\alpha g(s) k' \sin \theta}{2\sqrt{s}} \end{aligned}$$

$$\begin{aligned} T(1, 1; +, +) &= \frac{\alpha g(s) k' m_e (2 \cos \theta + 1)}{8s} \\ T(-1, -1; +, +) &= \frac{\alpha g(s) k' 2 \sin \theta - 1}{8s} \\ T(1, -1; +, -) &= -\frac{\alpha g(s) k' m_e (2 \sin \theta - 1)}{8s} \\ T(1, -1; +, +) &= -\frac{\alpha g(s) k' 2 \sin \theta}{4s} \\ T(1, 1; +, -) &= \frac{\alpha g(s) k' m_e (2 \cos \theta + 1)}{4s} \\ T(1, -1; -, +) &= -\frac{\alpha g(s) k' (1 - 2 \cos \theta)}{8\sqrt{s}} \end{aligned}$$

计算分波振幅, 首先变换到 $|J, M; \lambda_1, \lambda_2\rangle$ 表象

$$\begin{aligned}
\langle 0, 0|F^1|+, +\rangle &= 2\pi \int_{-1}^1 d(\cos \theta) d_{00}^1(\theta) T(0, 0; +, +) \\
&= 2\pi \int_{-1}^1 d(\cos \theta) \frac{\alpha g(s) k' \cos^2 \theta m_e}{s} \\
&= \frac{4\pi e g(s) k' m_e}{3s}
\end{aligned}$$

$$\begin{aligned}
\langle 0, 0|F^1|-, -\rangle &= 2\pi \int_{-1}^1 d(\cos \theta) d_{00}^1(\theta) T(0, 0; -, -) \\
&= 2\pi \int_{-1}^1 d(\cos \theta) d_{00}^1(\theta) T(0, 0; +, +) \\
&= \langle 0, 0|F^1|+, +\rangle
\end{aligned}$$

$$\begin{aligned}
\langle 0, 0|F^1|+, -\rangle &= 2\pi \int_{-1}^1 d(\cos \theta) d_{10}^1(\theta) T(0, 0; +, -) \\
&= 2\pi \int_{-1}^1 d(\cos \theta) \frac{\alpha g(s) k' \sin^2 \theta}{2\sqrt{2}s} \\
&= \frac{2\sqrt{2}\pi e g(s) k'}{3\sqrt{s}}
\end{aligned}$$

$$\begin{aligned}
\langle 0, 0|F^1|-, +\rangle &= 2\pi \int_{-1}^1 d(\cos \theta) d_{-10}^1(\theta) T(0, 0; -, +) \\
&= 2\pi \int_{-1}^1 d(\cos \theta) (-d_{10}^1(\theta)) (-T(0, 0; +, -)) \\
&= \langle 0, 0|F^1|+, -\rangle
\end{aligned}$$

因为考虑单光子交换，所以 $J = 1$ 。变换到 $[L, S, J]$ 表象，光子的量子数为 1^{--} ，所以 e^+e^- 系统在 3S_1 和 3D_1 分波，而 D^+D^- 在 1P_1 分波，首先计算矩阵元。对于 e^+e^- 系统 $L = 0$ 和 $S = 1$ 情形

$$\begin{aligned}
\langle 1, M; 0, 1|1, M; +, +\rangle &= \frac{1}{\sqrt{3}} \langle 0, 1; 0, 0|1, 0\rangle \langle \frac{1}{2}, \frac{1}{2}; \frac{1}{2}, -\frac{1}{2}|1, 0\rangle = \frac{1}{\sqrt{6}} \\
\langle 1, M; 0, 1|1, M; -, -\rangle &= \frac{1}{\sqrt{3}} \langle 0, 1; 0, 0|1, 0\rangle \langle \frac{1}{2}, \frac{1}{2}; -\frac{1}{2}, \frac{1}{2}|1, 0\rangle = \frac{1}{\sqrt{6}} \\
\langle 1, M; 0, 1|1, M; +, -\rangle &= \frac{1}{\sqrt{3}} \langle 0, 1; 0, 1|1, 1\rangle \langle \frac{1}{2}, \frac{1}{2}; \frac{1}{2}, -\frac{1}{2}|1, 1\rangle = \frac{1}{\sqrt{3}} \\
\langle 1, M; 0, 1|1, M; -, +\rangle &= \frac{1}{\sqrt{3}} \langle 0, 1; 0, -1|1, -1\rangle \langle \frac{1}{2}, \frac{1}{2}; -\frac{1}{2}, -\frac{1}{2}|1, -1\rangle = \frac{1}{\sqrt{3}}
\end{aligned}$$

对于 e^+e^- 系统 $L = 2$ 和 $S = 1$ 的情形

$$\begin{aligned}
\langle 1, M; 2, 1|1, M; +, +\rangle &= \sqrt{\frac{5}{3}} \langle 2, 1; 0, 0|1, 0\rangle \langle \frac{1}{2}, \frac{1}{2}; \frac{1}{2}, -\frac{1}{2}|1, 0\rangle = -\frac{1}{\sqrt{3}} \\
\langle 1, M; 2, 1|1, M; -, -\rangle &= \sqrt{\frac{5}{3}} \langle 2, 1; 0, 0|1, 0\rangle \langle \frac{1}{2}, \frac{1}{2}; -\frac{1}{2}, \frac{1}{2}|1, 0\rangle = -\frac{1}{\sqrt{3}} \\
\langle 1, M; 2, 1|1, M; +, -\rangle &= \sqrt{\frac{5}{3}} \langle 2, 1; 0, 1|1, 1\rangle \langle \frac{1}{2}, \frac{1}{2}; \frac{1}{2}, -\frac{1}{2}|1, 1\rangle = \frac{1}{\sqrt{6}}
\end{aligned}$$

$$\langle 1, M; 2, 1 | 1, M; -, + \rangle = \sqrt{\frac{5}{3}} \langle 2, 1; 0, -1 | 1, -1 \rangle \langle \frac{1}{2}, \frac{1}{2}; -\frac{1}{2}, -\frac{1}{2} | 1, -1 \rangle = \frac{1}{\sqrt{6}}$$

对于 $D\bar{D}$ 系统, $L = 1$ 和 $S = 0$ 的情形

$$\langle 1, M; 1, 0 | 1, M; 0, 0 \rangle = \langle 1, 0; 0, 0 | 1, 0 \rangle \langle 0, 0; 0, 0 | 0, 0 \rangle = 1$$

$$\begin{aligned} \langle {}^1P_1 | F | {}^3S_1 \rangle &= \sum_{\lambda_1, \lambda_2} \langle 1, M; 1, 0 | 1, M; 0, 0 \rangle \langle 0, 0 | F^1 | \lambda_1, \lambda_2 \rangle \langle 1, M; \lambda_1, \lambda_2 | 1, M; 0, 1 \rangle \\ &= \frac{1}{\sqrt{6}} (\langle 0, 0 | F^1 | +, + \rangle + \langle 0, 0 | F^1 | -, - \rangle) + \frac{1}{\sqrt{3}} (\langle 0, 0 | F^1 | +, - \rangle + \langle 0, 0 | F^1 | -, + \rangle) \\ &= \frac{8\pi\alpha g(s)k'm_e}{3\sqrt{6}s} + \frac{4\sqrt{2}\pi\alpha g(s)k'}{3\sqrt{3}\sqrt{s}} \\ &= \frac{2\sqrt{6}\pi\alpha g(s)}{9} \left(1 + \frac{m_e}{\sqrt{s}} \right) \frac{k'}{\sqrt{s}} \end{aligned}$$

$$\begin{aligned} \langle {}^1P_1 | F | {}^3D_1 \rangle &= \sum_{\lambda_1, \lambda_2} \langle 1, M; 1, 0 | 1, M; 0, 0 \rangle \langle 0, 0 | F^1 | \lambda_1, \lambda_2 \rangle \langle 1, M; \lambda_1, \lambda_2 | 1, M; 2, 1 \rangle \\ &= -\frac{1}{\sqrt{3}} (\langle 0, 0 | F^1 | +, + \rangle + \langle 0, 0 | F^1 | -, - \rangle) + \frac{1}{\sqrt{6}} (\langle 0, 0 | F^1 | +, - \rangle + \langle 0, 0 | F^1 | -, + \rangle) \\ &= -\frac{8\pi\alpha g(s)k'm_e}{3\sqrt{3}s} + \frac{4\pi\alpha g(s)k'}{3\sqrt{3}\sqrt{s}} \\ &= \frac{4\sqrt{6}\pi\alpha g(s)}{9} \frac{1}{\sqrt{2}} \left(1 - \frac{2m_e}{\sqrt{s}} \right) \frac{k'}{\sqrt{s}} \end{aligned}$$

其中 γe^+e^- 顶点

$$\begin{aligned} f_0^{e^+e^-} &= \left(1 + \frac{m_e}{\sqrt{s}} \right), \\ f_2^{e^+e^-} &= \frac{1}{\sqrt{2}} \left(1 - \frac{2m_e}{\sqrt{s}} \right) \end{aligned}$$

其中 γD^+D^- 顶点

$$f_1^{D\bar{D}} = \frac{g(s)k'}{\sqrt{s}}$$

螺旋度振幅与分波振幅的关系

$$\begin{aligned} \langle 0, 0 | F^1 | +, + \rangle &= \langle 0, 0 | F^1 | -, - \rangle = 4\pi \left(\frac{1}{\sqrt{6}} F_{10} - \frac{1}{\sqrt{3}} F_{12} \right) \\ \langle 0, 0 | F^1 | +, - \rangle &= \langle 0, 0 | F^1 | -, + \rangle = 4\pi \left(\frac{1}{\sqrt{3}} F_{10} + \frac{1}{\sqrt{6}} F_{12} \right) \end{aligned}$$

螺旋度振幅

$$\begin{aligned}
T(0, 0; \lambda_1, \lambda_2) &= \frac{3}{4\pi} d_{\lambda_0}^1(\theta) \langle 0, 0 | F^1 | \lambda_1, \lambda_2 \rangle \\
T(0, 0; +, +) &= \frac{3 \cos \theta}{4\pi} \langle 0, 0 | F^1 | +, + \rangle = 3 \cos \theta \left(\frac{1}{\sqrt{6}} F_{10} - \frac{1}{\sqrt{3}} F_{12} \right) \\
T(0, 0; -, -) &= T(0, 0; +, +) \\
T(0, 0; +, -) &= -\frac{3 \sin \theta}{4\sqrt{2}\pi} \langle 0, 0 | F^1 | +, - \rangle = -\frac{3 \sin \theta}{\sqrt{2}} \left(\frac{1}{\sqrt{3}} F_{10} + \frac{1}{\sqrt{6}} F_{12} \right) \\
T(0, 0; -, +) &= -T(0, 0; +, -)
\end{aligned}$$

振幅模方求和

$$\begin{aligned}
&\frac{1}{4} \sum_{\lambda_1, \lambda_2} |T(0, 0; \lambda_1, \lambda_2)|^2 \\
&= \frac{1}{2} (|T(0, 0; +, +)|^2 + |T(0, 0; +, -)|^2) \\
&= \frac{9 \cos^2 \theta}{2} \left(\frac{|F_{10}|^2}{6} + \frac{|F_{12}|^2}{3} - \frac{1}{3\sqrt{2}} (F_{10}^* F_{12} + F_{10} F_{12}^*) \right) \\
&\quad + \frac{9 \sin^2 \theta}{4} \left(\frac{|F_{10}|^2}{3} + \frac{|F_{12}|^2}{6} + \frac{1}{3\sqrt{2}} (F_{10}^* F_{12} + F_{10} F_{12}^*) \right) \\
&= \frac{3}{4} \left(|F_{10}|^2 + |F_{12}|^2 \left(2 \cos^2 \theta + \frac{\sin^2 \theta}{2} \right) + \frac{1}{\sqrt{2}} (F_{10}^* F_{12} + F_{10} F_{12}^*) (\sin^2 \theta - 2 \cos^2 \theta) \right)
\end{aligned}$$

在质心系中微分截面为, 其中 $\beta = k'/k$

$$\frac{d\sigma}{d\Omega} = \frac{\beta}{64\pi^2 s} \frac{1}{4} \sum |M|^2$$

最终

$$\sigma = \frac{3\pi\beta C(s)}{s} (|F_{10}|^2 + |F_{12}|^2)$$

通过 LS 方程解出末态 $D\bar{D}$ 重散射振幅 $T_{LL}(p, p'; E)$, 利用 DWBA 可以得到 $\gamma D\bar{D}$ 顶角

$$f_L^{D\bar{D}}(p) = f_L^{D\bar{D},0}(p) + \sum_{L'} \int_0^\infty \frac{dp'}{(2\pi)^3} \frac{p'^2}{f_L^{D\bar{D},0}(p')} \times \frac{1}{E - 2E_{p'} + i0^+} T_{LL}(p, p'; E)$$

其中 $\gamma D\bar{D}$ 裸顶点

$$f_L^{D\bar{D},0} = \frac{g_0(s)k'}{\sqrt{s}}$$

到 $D^* \bar{D}^*$ 有 $|^1P_1\rangle, |^5P_1\rangle, |^5F_1\rangle$

$$T(\pm 1, \pm 1; \lambda_1, \lambda_2) = N_i \frac{\alpha}{2} g(s) \bar{v}^e(-\mathbf{k}, \lambda_2) \gamma_\mu u^e(\mathbf{k}, \lambda_1) \frac{1}{s} [(p_3 - p_4)^\mu (\epsilon_3^\nu \cdot \epsilon_4^{\dagger\nu}) + \epsilon_3^\nu \cdot p_4^\nu \cdot \epsilon_4^{\dagger\mu} + \epsilon_4^{\dagger\nu} \cdot p_3^\nu \cdot \epsilon_3^\mu]$$

$$\begin{aligned}
T(1, 1; +, +) &= \frac{\alpha g(s) k' m_e (2 \cos \theta + 1)}{8s} \\
T(-1, -1; +, +) &= \frac{\alpha g(s) k' 2 \sin \theta - 1}{8s} \\
T(1, -1; +, -) &= -\frac{\alpha g(s) k' m_e (2 \sin \theta - 1)}{8s} \\
T(1, -1; +, +) &= -\frac{\alpha g(s) k' 2 \sin \theta}{4s} \\
T(1, 1; +, -) &= \frac{\alpha g(s) k' m_e (2 \cos \theta + 1)}{4s} \\
T(1, -1; -, +) &= -\frac{\alpha g(s) k' (1 - 2 \cos \theta)}{8\sqrt{s}}
\end{aligned}$$

$$\begin{aligned}
\langle {}^1P_1 | F | {}^3S_1 \rangle &= \sum_{\lambda_1, \lambda_2} \langle 1, M; 1, 1 | 1, M; 1, 1 \rangle \langle 1, 1 | F^1 | \lambda_1, \lambda_2 \rangle \langle 1, M; \lambda_1, \lambda_2 | 1, M; 0, 1 \rangle \\
&= \frac{2\sqrt{6}\pi\alpha g(s)}{24} \left(1 + \frac{m_e}{s}\right) \frac{k'}{s}
\end{aligned}$$

$$\begin{aligned}
\langle {}^5P_1 | F | {}^3S_1 \rangle &= \sum_{\lambda_1, \lambda_2} \langle 1, M; 1, 1 | 1, M; 1, 2 \rangle \langle 1, 1 | F^1 | \lambda_1, \lambda_2 \rangle \langle 1, M; \lambda_1, \lambda_2 | 1, M; 0, 1 \rangle \\
&= \frac{\pi\alpha g(s)}{12} \left(1 + \frac{m_e}{s}\right) \frac{k'}{s}
\end{aligned}$$

$$\begin{aligned}
\langle {}^1P_1 | F | {}^3D_1 \rangle &= \sum_{\lambda_1, \lambda_2} \langle 1, M; 1, 1 | 1, M; 1, 1 \rangle \langle 1, 1 | F^1 | \lambda_1, \lambda_2 \rangle \langle 1, M; \lambda_1, \lambda_2 | 1, M; 2, 1 \rangle \\
&= \frac{\sqrt{6}\pi\alpha g(s)}{32} \frac{1}{\sqrt{2}} \left(1 - \frac{2m_e}{s}\right) \frac{k'}{s}
\end{aligned}$$

$$\begin{aligned}
\langle {}^5F_1 | F | {}^3S_1 \rangle &= \sum_{\lambda_1, \lambda_2} \langle 1, M; 1, 1 | 1, M; 3, 2 \rangle \langle 1, 1 | F^1 | \lambda_1, \lambda_2 \rangle \langle 1, M; \lambda_1, \lambda_2 | 1, M; 0, 1 \rangle \\
&= \frac{\sqrt{6}\pi\alpha g(s)}{16} \left(1 + \frac{m_e}{s}\right) \frac{k'}{s}
\end{aligned}$$

$$\begin{aligned}
\langle {}^1F_1 | F | {}^3D_1 \rangle &= \sum_{\lambda_1, \lambda_2} \langle 1, M; 1, 1 | 1, M; 3, 2 \rangle \langle 1, 1 | F^1 | \lambda_1, \lambda_2 \rangle \langle 1, M; \lambda_1, \lambda_2 | 1, M; 2, 1 \rangle \\
&= \frac{\pi\alpha g(s)}{16} \frac{1}{\sqrt{2}} \left(1 - \frac{2m_e}{s}\right) \frac{k'}{s}
\end{aligned}$$

表 3: $D^* \bar{D}^* \text{I} = 0$ 的同位旋因子表

	中间态	$\pi\pi$	KK	$\eta\eta$	$\pi\eta$
足球图	-	-24	-16	-	-
左三角图	D	2	-	-	-
	D_s	-	2	-	-
右三角图	D	2	-	-	-
	D_s	-	2	-	-
平行箱图	$D\bar{D}$	$\frac{9}{16}$	-	$\frac{1}{144}$	$\frac{5}{48}$
	$D_s D_s$	-	$\frac{1}{2}$	-	-
交叉箱图	$D\bar{D}$	$\frac{1}{8}$	-	$\frac{1}{144}$	-
	$D_s D_s$	-	-	-	-

 表 4: $D\bar{D} \rightarrow D_s^+ D_s^- \text{I} = 0$ 的同位旋因子表

	中间态	πK	$K\eta$
足球图	-	$-12\sqrt{2}$	$-12\sqrt{2}$
左三角图	D^*	$\frac{3}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$
	D_s^*	-	$\sqrt{2}$
右三角图	D^*	$\frac{3}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$
	D_s^*	-	$\sqrt{2}$
平行箱图	$D^* \bar{D}^*$	$\frac{3}{4\sqrt{2}}$	$\frac{1}{12\sqrt{2}}$
	$D_s^* \bar{D}_s^*$	-	$\frac{1}{3\sqrt{2}}$

$$\int \frac{d^3 l}{(2\pi)^3} \frac{l^2 - q^2}{(\omega_+ + 2\Delta)(\omega_- + 2\Delta)(\omega_+ + \omega_-)} \quad (10)$$

由于

$$\begin{aligned} & \frac{1}{(\omega_+ + 2\Delta)(\omega_- + 2\Delta)(\omega_+ + \omega_-)} \\ &= \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{d\lambda}{(\omega_+^2 + \lambda^2)(\omega_-^2 + \lambda^2)(4\Delta^2 + \lambda^2)} \\ &= \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{d\lambda}{(\omega_+^2 + \lambda^2)(\omega_-^2 + \lambda^2)} - \frac{4\Delta^2}{\pi} \int_{-\infty}^{\infty} \frac{d\lambda}{(\omega_+^2 + \lambda^2)(\omega_-^2 + \lambda^2)(4\Delta^2 + \lambda^2)} \end{aligned} \quad (11)$$

其中第一项是忽略质量差的结果，第二项是质量差不为零所带来的修正。现在针对第二项进行详细计算

$$-8\Delta^2 \int_{-\infty}^{\infty} \frac{d\lambda}{2\pi} \int \frac{d^3 l}{(2\pi)^3} \frac{l^2 - q^2}{(\omega_+^2 + \lambda^2)(\omega_-^2 + \lambda^2)(4\Delta^2 + \lambda^2)} \quad (12)$$

首先考虑如下积分

$$\int_{-\infty}^{\infty} \frac{d\lambda}{2\pi} \int \frac{d^3 l}{(2\pi)^3} \frac{1}{((l+q)^2 + 4m_1^2 + \lambda^2)((l-q)^2 + 4m_2^2 + \lambda^2)(4\Delta^2 + \lambda^2)}$$

通过费曼参数化公式 $\frac{1}{ABC} = \int_0^{\infty} dy \int_0^1 dx \frac{4}{[2Ay + Bx + (1-x)C]^3}$ (13)

$$= \int_0^{\infty} dy \int_0^1 dx \int_{-\infty}^{\infty} \frac{d\lambda}{2\pi} \int \frac{d^3 l}{(2\pi)^3} \frac{4}{[2y(4\Delta^2 + \lambda^2) + x((l+q)^2 + 4m_1^2 + \lambda^2) + (1-x)((l-q)^2 + 4m_2^2 + \lambda^2)]^3}$$

$$= \int_0^{\infty} dy \int_0^1 dx \int_{-\infty}^{\infty} \frac{d\lambda}{2\pi} \int \frac{d^3 l}{(2\pi)^3} \frac{4}{[(l + (2x-1)q)^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + 8y\Delta^2 + (1+2y)\lambda^2]^3}$$

利用平移不变性 (14)

$$= \int_0^{\infty} dy \int_0^1 dx \int_{-\infty}^{\infty} \frac{d\lambda}{2\pi} \int \frac{d^3 l}{(2\pi)^3} \frac{4}{[l^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + 8y\Delta^2 + (1+2y)\lambda^2]^3} \quad (15)$$

对 λ 进行积分变换 $\sqrt{1+2y}\lambda \rightarrow \lambda$, 那么积分变为

$$\int_0^{\infty} \frac{dy}{\sqrt{2y+1}} \int_0^1 dx \int_{-\infty}^{\infty} \frac{d\lambda}{2\pi} \int \frac{d^3 l}{(2\pi)^3} [l^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + 8y\Delta^2 + \lambda^2]^3 \quad (16)$$

定义如下欧式空间四矢量 $k^\mu = (k^0, k) = (\lambda, l)$, 其模方为 $k^2 = \lambda^2 + l^2$ 。积分变为

$$\int_0^{\infty} \frac{dy}{\sqrt{2y+1}} \int_0^1 dx \int \frac{d^4 k}{(2\pi)^4} [k^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + 8y\Delta^2]^3 \quad (17)$$

将四维积分变为 d 维

$$\mu^{4-d} \int \frac{d^d k}{(2\pi)^d} \frac{1}{[k^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + 8y\Delta^2]^3}$$

$$= \frac{2\mu^{4-d}}{(4\pi)^{d/2}\Gamma(\frac{d}{2})} \int_0^{\infty} \frac{k^{d-1} dk}{[k^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + 8y\Delta^2]^3}$$

$$= \frac{\mu^{4-d}}{2(4\pi)^{d/2}[4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + 8y\Delta^2]^{3-\frac{d}{2}}} \quad (18)$$

取 $d = 4 - \epsilon$, 令 $\epsilon \rightarrow 0$

$$= \frac{\mu^\epsilon}{2(4\pi)^{2-\epsilon/2}[4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + 8y\Delta^2]^{1+\frac{\epsilon}{2}}}$$

$$= \frac{1}{32\pi^2[4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + 8y\Delta^2]} + O(\epsilon^1) \quad (19)$$

得到圈积分为

$$\int_0^{\infty} \frac{dy}{\sqrt{2y+1}} \int_0^1 dx \frac{1}{8\pi^2[4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + 8y\Delta^2]}$$

$$= \int_0^1 dx \frac{\arccos\left(\frac{2\Delta}{\sqrt{x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2}}\right)}{16\pi^2\Delta\sqrt{4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 - 4\Delta^2}} \quad (20)$$

第二个标量积分为

$$\int_{-\infty}^{\infty} \frac{d\lambda}{2\pi} \int \frac{d^3 l}{(2\pi)^3} \frac{l^2}{((l+q)^2 + 4m_1^2 + \lambda^2)((l-q)^2 + 4m_2^2 + \lambda^2)(4\Delta^2 + \lambda^2)}$$

$$= \int_0^{\infty} dy \int_0^1 dx \int_{-\infty}^{\infty} \frac{d\lambda}{2\pi} \int \frac{d^3 l}{(2\pi)^3} \frac{4[l^2 - 2(2x-1)l \cdot q + (2x-1)^2 q^2]}{[l^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + 8y\Delta^2 + (1+2y)\lambda^2]^3}$$

$$= \int_0^{\infty} dy \int_0^1 dx \int_{-\infty}^{\infty} \frac{d\lambda}{2\pi} \int \frac{d^3 l}{(2\pi)^3} \frac{4[l^2 + (2x-1)^2 q^2]}{[l^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + 8y\Delta^2 + (1+2y)\lambda^2]^3} \quad (21)$$

其中 l 的奇次幂对积分贡献为零。正比于 l^2 项为

$$\int_0^\infty \frac{dy}{\sqrt{2y+1}} \int_0^1 dx \int \frac{d^4 k}{(2\pi)^4} \frac{l^2}{[k^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + 8y\Delta^2]^3} \quad (22)$$

拓展到 d 维，对于张量积分有

$$\begin{aligned} & \int \frac{d^d k}{(2\pi)^d} \frac{k^\mu k^\nu}{[k^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + 8y\Delta^2]^3} \\ &= \frac{g^{\mu\nu}}{d} \int \frac{d^d k}{(2\pi)^d} \frac{k^2}{[k^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + 8y\Delta^2]^3} \end{aligned} \quad (23)$$

$l^2 = k^i k^j g_{ij}$ ，这里指标 i, j 为三动量空间的指标。那么

$$\begin{aligned} & \int \frac{d^d k}{(2\pi)^d} \frac{k^i k^j g_{ij}}{[k^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + 8y\Delta^2]^3} \\ &= \frac{g_{ij} g^{ij}}{d} \int \frac{d^d k}{(2\pi)^d} \frac{k^2}{[k^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + 8y\Delta^2]^3} \\ &= \frac{d-1}{d} \int \frac{d^d k}{(2\pi)^d} \frac{k^2}{[k^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + 8y\Delta^2]^3} \end{aligned} \quad (24)$$

$$\begin{aligned} & \frac{d-1}{d} \mu^{4-d} \int \frac{d^d k}{(2\pi)^d} \frac{k^2}{[k^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + 8y\Delta^2]^3} \\ &= \frac{d-1}{d} \frac{2\mu^{4-d}}{(4\pi)^{d/2} \Gamma(\frac{d}{2})} \int_0^\infty \frac{k^{d+1} dk}{[k^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + 8y\Delta^2]^3} \\ &= \frac{d-1}{d} \frac{2\mu^{4-d}}{(4\pi)^{d/2} \Gamma(\frac{d}{2})} \frac{\Gamma(1 + \frac{d}{2}) \Gamma(2 - \frac{d}{2})}{2\Gamma(3) [4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + 8y\Delta^2]^{2-\frac{d}{2}}} \\ &= \frac{d-1}{4} \frac{\mu^{4-d}}{(4\pi)^{d/2}} \frac{\Gamma(2 - \frac{d}{2})}{[4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + 8y\Delta^2]^{2-\frac{d}{2}}} \end{aligned} \quad (25)$$

取 $d = 4 - \epsilon$ 并令 $\epsilon \rightarrow 0$

$$\begin{aligned} & \frac{3-\epsilon}{4} \frac{\mu^\epsilon}{(4\pi)^{2-\epsilon/2}} \frac{\Gamma(\frac{\epsilon}{2})}{[4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + 8y\Delta^2]^{\frac{\epsilon}{2}}} \\ &= \frac{3}{64\pi^2} R - \frac{3}{64\pi^2} \left[\ln \left(\frac{4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + 8y\Delta^2}{\mu^2} \right) + \frac{5}{3} \right] \end{aligned} \quad (26)$$

其中

$$R = -\frac{2}{\epsilon} + \gamma_E - \ln(4\pi) - 1 = \frac{2}{d-4} + \gamma_E - \ln(4\pi) - 1 \quad (27)$$

所以正比于 l^2 的积分为

$$\int_0^\infty \frac{dy}{\sqrt{2y+1}} \int_0^1 dx \left(\frac{3}{64\pi^2} R - \frac{3}{64\pi^2} \left[\ln \left(\frac{4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + 8y\Delta^2}{\mu^2} \right) + \frac{5}{3} \right] \right) \quad (28)$$

同样积分类型

$$\int_{-\infty}^\infty \frac{d\lambda}{2\pi} \int \frac{d^3 l}{(2\pi)^3} \frac{1}{((l+q)^2 + 4m_1^2 + \lambda^2)((l-q)^2 + 4m_2^2 + \lambda^2)(4\Delta^2 + \lambda^2)}$$

$$\text{通过费曼参数化公式 } \frac{1}{ABC} = \int_0^{1-x} dy \int_0^1 dx \frac{2}{[Ax + By + (1-x-y)C]^3} \quad (29)$$

$$\begin{aligned} &= \int_0^{1-x} dy \int_0^1 dx \int_{-\infty}^\infty \frac{d\lambda}{2\pi} \int \frac{d^3 l}{(2\pi)^3} \frac{2}{[y((l+q)^2 + 4m_1^2 + \lambda^2) + x((l-q)^2 + 4m_2^2 + \lambda^2) + (1-x-y)(4\Delta^2 + \lambda^2)]^3} \\ &= \int_0^{1-x} dy \int_0^1 dx \int_{-\infty}^\infty \frac{d\lambda}{2\pi} \int \frac{d^3 l}{(2\pi)^3} \frac{2}{\left[(y+x)(l + \frac{y-x}{y+x}q)^2 + \frac{4xy}{y+x}q^2 + 4xm_2^2 + 4ym_1^2 + 4(1-x-y)\Delta^2 + \lambda^2 \right]^3} \end{aligned}$$

对 λ 进行积分变换 $\sqrt{x+y}\lambda \rightarrow \lambda$, 那么积分变为

$$\int_0^{1-x} \frac{dy}{(x+y)^{\frac{5}{2}}} \int_0^1 dx \int_{-\infty}^{\infty} \frac{d\lambda}{2\pi} \int \frac{d^3 l}{(2\pi)^3} \frac{2}{\left[l^2 + \frac{4xy}{(x+y)^2} q^2 + \frac{4xm_2^2}{x+y} + \frac{4ym_1^2}{x+y} + \frac{4(1-x-y)\Delta^2}{x+y} + \lambda^2\right]^3} \quad (30)$$

定义如下欧式空间四矢量 $k^\mu = (k^0, k) = (\lambda, l)$, 其模方为 $k^2 = \lambda^2 + l^2$. 积分变为

$$\int_0^{1-x} \frac{dy}{(x+y)^{\frac{5}{2}}} \int_0^1 dx \int \frac{d^4 k}{(2\pi)^4} \frac{2}{\left[k^2 + \frac{4xy}{(x+y)^2} q^2 + \frac{4xm_2^2}{x+y} + \frac{4ym_1^2}{x+y} + \frac{4(1-x-y)\Delta^2}{x+y}\right]^3} \quad (31)$$

将四维积分变为 d 维

$$\begin{aligned} & \mu^{4-d} \int \frac{d^d k}{(2\pi)^d} \frac{1}{\left[k^2 + \frac{4xy}{(x+y)^2} q^2 + \frac{4xm_2^2}{x+y} + \frac{4ym_1^2}{x+y} + \frac{4(1-x-y)\Delta^2}{x+y}\right]^3} \\ &= \frac{2\mu^{4-d}}{(4\pi)^{d/2} \Gamma(\frac{d}{2})} \int_0^\infty \frac{k^{d-1} dk}{\left[k^2 + \frac{4xy}{(x+y)^2} q^2 + \frac{4xm_2^2}{x+y} + \frac{4ym_1^2}{x+y} + \frac{4(1-x-y)\Delta^2}{x+y}\right]^3} \\ &= \frac{\mu^{4-d}}{2(4\pi)^{d/2} \left[\frac{4xy}{(x+y)^2} q^2 + \frac{4xm_2^2}{x+y} + \frac{4ym_1^2}{x+y} + \frac{4(1-x-y)\Delta^2}{x+y}\right]^{3-\frac{d}{2}}} \end{aligned} \quad (32)$$

取 $d = 4 - \epsilon$, 令 $\epsilon \rightarrow 0$

$$\begin{aligned} &= \frac{\mu^\epsilon}{2(4\pi)^{2-\epsilon/2} \left[\frac{4xy}{(x+y)^2} q^2 + \frac{4xm_2^2}{x+y} + \frac{4ym_1^2}{x+y} + \frac{4(1-x-y)\Delta^2}{x+y}\right]^{1+\frac{\epsilon}{2}}} \\ &= \frac{1}{32\pi^2 \left[\frac{4xy}{(x+y)^2} q^2 + \frac{4xm_2^2}{x+y} + \frac{4ym_1^2}{x+y} + \frac{4(1-x-y)\Delta^2}{x+y}\right]} + O(\epsilon^1) \end{aligned} \quad (33)$$

得到圈积分为

$$\begin{aligned} & \int_0^{1-x} \frac{dy}{(x+y)^{\frac{5}{2}}} \int_0^1 dx \frac{1}{16\pi^2 \left[\frac{4xy}{(x+y)^2} q^2 + \frac{4xm_2^2}{x+y} + \frac{4ym_1^2}{x+y} + \frac{4(1-x-y)\Delta^2}{x+y}\right]} \\ &= \int_0^1 dx \frac{\arccos\left(\frac{2\Delta}{\sqrt{x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 - 4\Delta^2}}\right)}{16\pi^2 \Delta \sqrt{4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 - 4\Delta^2}} \end{aligned} \quad (34)$$

第二个标量积分为

$$\begin{aligned} & \int_{-\infty}^{\infty} \frac{d\lambda}{2\pi} \int \frac{d^3 l}{(2\pi)^3} \frac{l^2}{((l+q)^2 + 4m_1^2 + \lambda^2)((l-q)^2 + 4m_1^2 + \lambda^2)(4\Delta^2 + \lambda^2)} \\ &= \int_0^{1-x} \frac{dy}{(x+y)^{\frac{5}{2}}} \int_0^1 dx \int_{-\infty}^{\infty} \frac{d\lambda}{2\pi} \int \frac{d^3 l}{(2\pi)^3} \frac{2[l^2 + 2\frac{y-x}{y+x} l \cdot q + (\frac{y-x}{y+x})^2 q^2]}{\left[l^2 + \frac{4xy}{(x+y)^2} q^2 + \frac{4xm_2^2}{x+y} + \frac{4ym_1^2}{x+y} + \frac{4(1-x-y)\Delta^2}{x+y} + \lambda^2\right]^3} \\ &= \int_0^{1-x} \frac{dy}{(x+y)^{\frac{5}{2}}} \int_0^1 dx \int_{-\infty}^{\infty} \frac{d\lambda}{2\pi} \int \frac{d^3 l}{(2\pi)^3} \frac{2[l^2 + (\frac{y-x}{y+x})^2 q^2]}{\left[l^2 + \frac{4xy}{(x+y)^2} q^2 + \frac{4xm_2^2}{x+y} + \frac{4ym_1^2}{x+y} + \frac{4(1-x-y)\Delta^2}{x+y} + \lambda^2\right]^3} \end{aligned} \quad (35)$$

其中 l 的奇次幂对积分贡献为零。正比于 l^2 项为

$$\int_0^{1-x} \frac{dy}{(x+y)^{\frac{5}{2}}} \int_0^1 dx \int_{-\infty}^{\infty} \frac{d\lambda}{2\pi} \int \frac{d^3 l}{(2\pi)^3} \frac{2l^2}{\left[l^2 + \frac{4xy}{(x+y)^2} q^2 + \frac{4xm_2^2}{x+y} + \frac{4ym_1^2}{x+y} + \frac{4(1-x-y)\Delta^2}{x+y} + \lambda^2\right]^3} \quad (36)$$

拓展到 d 维，对于张量积分有

$$\begin{aligned}
& \int \frac{d^d k}{(2\pi)^d} \frac{k^\mu k^\nu}{\left[k^2 + \frac{4xy}{(x+y)^2} q^2 + \frac{4xm_2^2}{x+y} + \frac{4ym_1^2}{x+y} + \frac{4(1-x-y)\Delta^2}{x+y} \right]^3} \\
&= \frac{g^{\mu\nu}}{d} \int \frac{d^d k}{(2\pi)^d} \frac{k^2}{\left[k^2 + \frac{4xy}{(x+y)^2} q^2 + \frac{4xm_2^2}{x+y} + \frac{4ym_1^2}{x+y} + \frac{4(1-x-y)\Delta^2}{x+y} \right]^3}
\end{aligned} \tag{37}$$

$l^2 = k^i k^j g_{ij}$ ，这里指标 i, j 为三动量空间的指标。那么

$$\begin{aligned}
& \int \frac{d^d k}{(2\pi)^d} \frac{k^i k^j g_{ij}}{\left[k^2 + \frac{4xy}{(x+y)^2} q^2 + \frac{4xm_2^2}{x+y} + \frac{4ym_1^2}{x+y} + \frac{4(1-x-y)\Delta^2}{x+y} \right]^3} \\
&= \frac{g_{ij} g^{ij}}{d} \int \frac{d^d k}{(2\pi)^d} \frac{k^2}{\left[k^2 + \frac{4xy}{(x+y)^2} q^2 + \frac{4xm_2^2}{x+y} + \frac{4ym_1^2}{x+y} + \frac{4(1-x-y)\Delta^2}{x+y} \right]^3} \\
&= \frac{d-1}{d} \int \frac{d^d k}{(2\pi)^d} \frac{k^2}{\left[k^2 + \frac{4xy}{(x+y)^2} q^2 + \frac{4xm_2^2}{x+y} + \frac{4ym_1^2}{x+y} + \frac{4(1-x-y)\Delta^2}{x+y} \right]^3}
\end{aligned} \tag{38}$$

$$\begin{aligned}
& \frac{d-1}{d} \mu^{4-d} \int \frac{d^d k}{(2\pi)^d} \frac{k^2}{\left[k^2 + \frac{4xy}{(x+y)^2} q^2 + \frac{4xm_2^2}{x+y} + \frac{4ym_1^2}{x+y} + \frac{4(1-x-y)\Delta^2}{x+y} \right]^3} \\
&= \frac{d-1}{d} \frac{2\mu^{4-d}}{(4\pi)^{d/2} \Gamma(\frac{d}{2})} \int_0^\infty \frac{k^{d+1} dk}{\left[k^2 + \frac{4xy}{(x+y)^2} q^2 + \frac{4xm_2^2}{x+y} + \frac{4ym_1^2}{x+y} + \frac{4(1-x-y)\Delta^2}{x+y} \right]^3} \\
&= \frac{d-1}{d} \frac{2\mu^{4-d}}{(4\pi)^{d/2} \Gamma(\frac{d}{2})} \frac{\Gamma(1+\frac{d}{2}) \Gamma(2-\frac{d}{2})}{2\Gamma(3) \left[\frac{4xy}{(x+y)^2} q^2 + \frac{4xm_2^2}{x+y} + \frac{4ym_1^2}{x+y} + \frac{4(1-x-y)\Delta^2}{x+y} \right]^{2-\frac{d}{2}}} \\
&= \frac{d-1}{4} \frac{\mu^{4-d}}{(4\pi)^{d/2}} \frac{\Gamma(2-\frac{d}{2})}{\left[\frac{4xy}{(x+y)^2} q^2 + \frac{4xm_2^2}{x+y} + \frac{4ym_1^2}{x+y} + \frac{4(1-x-y)\Delta^2}{x+y} \right]^{2-\frac{d}{2}}}
\end{aligned} \tag{39}$$

取 $d = 4 - \epsilon$ 并令 $\epsilon \rightarrow 0$

$$\begin{aligned}
& \frac{3-\epsilon}{4} \frac{\mu^\epsilon}{(4\pi)^{2-\epsilon/2}} \frac{\Gamma(\frac{\epsilon}{2})}{\left[\frac{4xy}{(x+y)^2} q^2 + \frac{4xm_2^2}{x+y} + \frac{4ym_1^2}{x+y} + \frac{4(1-x-y)\Delta^2}{x+y} \right]^{\frac{\epsilon}{2}}} \\
&= R + \left[\ln \left(\frac{k^2 + \frac{4xy}{x+y} q^2 + 4(m_1^2 - m_2^2)x + 4ym_2^2 + (1-x-y)4\Delta^2}{\mu^2} \right) + \frac{5}{3} - \ln(x+y) \right]
\end{aligned} \tag{40}$$

其中

$$R = -\frac{2}{\epsilon} + \gamma_E - \ln(4\pi) - 1 = \frac{2}{d-4} + \gamma_E - \ln(4\pi) - 1 \tag{41}$$

所以正比于 l^2 的积分为

$$\int_0^{1-x} \frac{dy}{(x+y)^{\frac{5}{2}}} \int_0^1 dx \left[R + \left[\ln \left(\frac{k^2 + \frac{4xy}{x+y} q^2 + 4(m_1^2 - m_2^2)x + 4ym_2^2 + (1-x-y)4\Delta^2}{\mu^2} \right) + \frac{5}{3} - \ln(x+y) \right] \right] \tag{42}$$

第三种积分

$$\begin{aligned}
& \frac{\epsilon^i \cdot \epsilon^{\dagger j} g^{ij}}{3} \int_{-\infty}^{\infty} \frac{d\lambda}{2\pi} \int \frac{d^3 l}{(2\pi)^3} \frac{l^i l^j}{((l+q)^2 + 4m_1^2 + \lambda^2)((l-q)^2 + 4m_1^2 + \lambda^2)(4\Delta^2 + \lambda^2)} \\
&= \frac{\epsilon^i \cdot \epsilon^{\dagger j} g^{ij}}{3} \int_0^{1-x} \frac{dy}{(x+y)^{\frac{5}{2}}} \int_0^1 dx \int_{-\infty}^{\infty} \frac{d\lambda}{2\pi} \int \frac{d^3 l}{(2\pi)^3} \frac{2[l^2 + 2\frac{y-x}{y+x}l \cdot q + (\frac{y-x}{y+x})^2 q^2]}{\left[l^2 + \frac{4xy}{(x+y)^2}q^2 + \frac{4xm_2^2}{x+y} + \frac{4ym_1^2}{x+y} + \frac{4(1-x-y)\Delta^2}{x+y} + \lambda^2\right]^3} \\
&= \frac{\epsilon^i \cdot \epsilon^{\dagger j} g^{ij}}{3} \int_0^{1-x} \frac{dy}{(x+y)^{\frac{5}{2}}} \int_0^1 dx \int_{-\infty}^{\infty} \frac{d\lambda}{2\pi} \int \frac{d^3 l}{(2\pi)^3} \frac{2[l^2 + (\frac{y-x}{y+x})^2 q^2]}{\left[l^2 + \frac{4xy}{(x+y)^2}q^2 + \frac{4xm_2^2}{x+y} + \frac{4ym_1^2}{x+y} + \frac{4(1-x-y)\Delta^2}{x+y} + \lambda^2\right]^3} \quad (43)
\end{aligned}$$

其中 l 的奇次幂对积分贡献为零。正比于 l^2 项为

$$\frac{\epsilon^i \cdot \epsilon^{\dagger j} g^{ij}}{3} \int_0^{1-x} \frac{dy}{(x+y)^{\frac{5}{2}}} \int_0^1 dx \int_{-\infty}^{\infty} \frac{d\lambda}{2\pi} \int \frac{d^3 l}{(2\pi)^3} \frac{2l^2}{\left[l^2 + \frac{4xy}{(x+y)^2}q^2 + \frac{4xm_2^2}{x+y} + \frac{4ym_1^2}{x+y} + \frac{4(1-x-y)\Delta^2}{x+y} + \lambda^2\right]^3} \quad (44)$$

拓展到 d 维，对于张量积分有

$$\begin{aligned}
& \frac{\epsilon^i \cdot \epsilon^{\dagger j} g^{ij}}{3} \int \frac{d^d k}{(2\pi)^d} \frac{k^\mu k^\nu}{\left[k^2 + \frac{4xy}{(x+y)^2}q^2 + \frac{4xm_2^2}{x+y} + \frac{4ym_1^2}{x+y} + \frac{4(1-x-y)\Delta^2}{x+y}\right]^3} \\
&= \frac{\epsilon^i \cdot \epsilon^{\dagger j} g^{ij}}{3} \frac{g^{\mu\nu}}{d} \int \frac{d^d k}{(2\pi)^d} \frac{k^2}{\left[k^2 + \frac{4xy}{(x+y)^2}q^2 + \frac{4xm_2^2}{x+y} + \frac{4ym_1^2}{x+y} + \frac{4(1-x-y)\Delta^2}{x+y}\right]^3} \quad (45)
\end{aligned}$$

$l^2 = k^i k^j g_{ij}$ ，这里指标 i, j 为三动量空间的指标。那么

$$\begin{aligned}
& \frac{\epsilon^i \cdot \epsilon^{\dagger j} g^{ij}}{3} \int \frac{d^d k}{(2\pi)^d} \frac{k^i k^j g_{ij}}{\left[k^2 + \frac{4xy}{(x+y)^2}q^2 + \frac{4xm_2^2}{x+y} + \frac{4ym_1^2}{x+y} + \frac{4(1-x-y)\Delta^2}{x+y}\right]^3} \\
&= \frac{\epsilon^i \cdot \epsilon^{\dagger j} g^{ij}}{3} \frac{g_{ij} g^{ij}}{d} \int \frac{d^d k}{(2\pi)^d} \frac{k^2}{\left[k^2 + \frac{4xy}{(x+y)^2}q^2 + \frac{4xm_2^2}{x+y} + \frac{4ym_1^2}{x+y} + \frac{4(1-x-y)\Delta^2}{x+y}\right]^3} \\
&= \frac{\epsilon^i \cdot \epsilon^{\dagger j} g^{ij}}{3} \frac{d-1}{d} \int \frac{d^d k}{(2\pi)^d} \frac{k^2}{\left[k^2 + \frac{4xy}{(x+y)^2}q^2 + \frac{4xm_2^2}{x+y} + \frac{4ym_1^2}{x+y} + \frac{4(1-x-y)\Delta^2}{x+y}\right]^3} \quad (46) \\
& \frac{\epsilon^i \cdot \epsilon^{\dagger j} g^{ij}}{3} \frac{d-1}{d} \mu^{4-d} \int \frac{d^d k}{(2\pi)^d} \frac{k^2}{\left[k^2 + \frac{4xy}{(x+y)^2}q^2 + \frac{4xm_2^2}{x+y} + \frac{4ym_1^2}{x+y} + \frac{4(1-x-y)\Delta^2}{x+y}\right]^3} \\
&= \frac{\epsilon^i \cdot \epsilon^{\dagger j} g^{ij}}{3} \frac{d-1}{d} \frac{2\mu^{4-d}}{(4\pi)^{d/2} \Gamma(\frac{d}{2})} \int_0^\infty \frac{k^{d+1} dk}{\left[k^2 + \frac{4xy}{(x+y)^2}q^2 + \frac{4xm_2^2}{x+y} + \frac{4ym_1^2}{x+y} + \frac{4(1-x-y)\Delta^2}{x+y}\right]^3} \\
&= \frac{\epsilon^i \cdot \epsilon^{\dagger j} g^{ij}}{3} \frac{d-1}{d} \frac{2\mu^{4-d}}{(4\pi)^{d/2} \Gamma(\frac{d}{2})} \frac{\Gamma(1 + \frac{d}{2}) \Gamma(2 - \frac{d}{2})}{2\Gamma(3) \left[\frac{4xy}{(x+y)^2}q^2 + \frac{4xm_2^2}{x+y} + \frac{4ym_1^2}{x+y} + \frac{4(1-x-y)\Delta^2}{x+y}\right]^{2-\frac{d}{2}}} \\
&= \frac{\epsilon^i \cdot \epsilon^{\dagger j} g^{ij}}{3} \frac{d-1}{4} \frac{\mu^{4-d}}{(4\pi)^{d/2}} \frac{\Gamma(2 - \frac{d}{2})}{\left[\frac{4xy}{(x+y)^2}q^2 + \frac{4xm_2^2}{x+y} + \frac{4ym_1^2}{x+y} + \frac{4(1-x-y)\Delta^2}{x+y}\right]^{2-\frac{d}{2}}} \quad (47)
\end{aligned}$$

取 $d = 4 - \epsilon$ 并令 $\epsilon \rightarrow 0$

$$\begin{aligned}
& \frac{\epsilon^i \cdot \epsilon^{\dagger j} g^{ij}}{3} \frac{3-\epsilon}{4} \frac{\mu^\epsilon}{(4\pi)^{2-\epsilon/2}} \frac{\Gamma(\frac{\epsilon}{2})}{\left[\frac{4xy}{(x+y)^2}q^2 + \frac{4xm_2^2}{x+y} + \frac{4ym_1^2}{x+y} + \frac{4(1-x-y)\Delta^2}{x+y}\right]^{\frac{\epsilon}{2}}} \\
&= \frac{\epsilon^i \cdot \epsilon^{\dagger j} g^{ij}}{3} \left[R + \left[\ln \left(\frac{k^2 + \frac{4xy}{x+y}q^2 + 4(m_1^2 - m_2^2)x + 4ym_2^2 + (1-x-y)4\Delta^2}{\mu^2} \right) + \frac{5}{3} - \ln(x+y) \right] \right] \quad (48)
\end{aligned}$$

其中

$$R = -\frac{2}{\epsilon} + \gamma_E - \ln(4\pi) - 1 = \frac{2}{d-4} + \gamma_E - \ln(4\pi) - 1 \quad (49)$$

$$\frac{\epsilon^i \cdot \epsilon^{\dagger j} g^{ij}}{3} \int_0^{1-x} \frac{dy}{(x+y)^{\frac{5}{2}}} \int_0^1 dx [R + \left[\ln \left(\frac{k^2 + \frac{4xy}{x+y} q^2 + 4(m_1^2 - m_2^2)x + 4ym_2^2 + (1-x-y)4\Delta^2}{\mu^2} \right) + \frac{5}{3} - \ln(x+y) \right]] \quad (50)$$

Box 图的圈积分类型

$$\int \frac{d^3 l}{(2\pi)^3} \frac{(l^2 - q^2)^2}{(\omega_+ + 2\Delta)^2 (\omega_- + 2\Delta)^2 (\omega_+ + \omega_-)}$$

$$\int \frac{d^3 l}{(2\pi)^3} \frac{((\omega_+ + 2\Delta)^2 + (\omega_+ + 2\Delta)(\omega_- + 2\Delta) + (\omega_- + 2\Delta)^2)(l^2 - q^2)^2}{(\omega_+ + 2\Delta)^3 (\omega_- + 2\Delta)^3 (\omega_+ + \omega_-)} \quad (51)$$

推导公式：

$$\int_0^\infty \frac{x^\beta dx}{(x^2 + m^2)^\alpha} = \frac{\Gamma(\frac{\beta+1}{2}) \Gamma(\alpha - \frac{\beta+1}{2})}{2 \Gamma(\alpha) (m^2)^{\alpha - \frac{\beta+1}{2}}},$$

令 $t = x^2$ ，则 $dt = 2x dx$ ，且 $x^\beta = t^{\beta/2}$ ， $dx = dt/(2\sqrt{t})$ 。于是

$$\int_0^\infty \frac{x^\beta dx}{(x^2 + m^2)^\alpha} = \int_0^\infty \frac{t^{\beta/2}}{(t + m^2)^\alpha} \cdot \frac{dt}{2\sqrt{t}} = \frac{1}{2} \int_0^\infty \frac{t^{(\beta-1)/2}}{(t + m^2)^\alpha} dt.$$

再令 $t = m^2 y$ ，此时 $dt = m^2 dy$ ，且

$$t^{(\beta-1)/2} = (m^2 y)^{(\beta-1)/2} = m^{\beta-1} y^{(\beta-1)/2},$$

$$t + m^2 = m^2(1 + y) \Rightarrow (t + m^2)^{-\alpha} = m^{-2\alpha}(1 + y)^{-\alpha}.$$

代入得：

$$\frac{1}{2} \int_0^\infty \frac{t^{(\beta-1)/2}}{(t + m^2)^\alpha} dt = \frac{1}{2} \int_0^\infty \frac{m^{\beta-1} y^{(\beta-1)/2}}{m^{2\alpha}(1 + y)^\alpha} \cdot m^2 dy.$$

系数为

$$\frac{1}{2} \cdot \frac{m^{\beta+1}}{m^{2\alpha}} = \frac{1}{2} m^{\beta+1-2\alpha}.$$

于是积分为

$$\frac{1}{2} m^{\beta+1-2\alpha} \int_0^\infty \frac{y^{(\beta-1)/2}}{(1 + y)^\alpha} dy.$$

Beta 函数定义：

$$B(p, q) = \int_0^\infty \frac{y^{p-1}}{(1 + y)^{p+q}} dy = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}, \quad p > 0, q > 0.$$

设 $p = \frac{\beta-1}{2} + 1 = \frac{\beta+1}{2}$ ，且要求 $\alpha = p + q$ ，即

$$q = \alpha - p = \alpha - \frac{\beta+1}{2}.$$

此时被积函数为 $y^{p-1}/(1 + y)^{p+q} = y^{(\beta-1)/2}/(1 + y)^\alpha$ ，因此

$$\int_0^\infty \frac{y^{(\beta-1)/2}}{(1 + y)^\alpha} dy = B\left(\frac{\beta+1}{2}, \alpha - \frac{\beta+1}{2}\right) = \frac{\Gamma(\frac{\beta+1}{2}) \Gamma(\alpha - \frac{\beta+1}{2})}{\Gamma(\alpha)}.$$

原积分变为

$$\frac{1}{2} m^{\beta+1-2\alpha} \cdot \frac{\Gamma(\frac{\beta+1}{2}) \Gamma(\alpha - \frac{\beta+1}{2})}{\Gamma(\alpha)}.$$

注意 $\beta + 1 - 2\alpha = -2\left(\alpha - \frac{\beta+1}{2}\right)$, 所以

$$m^{\beta+1-2\alpha} = m^{-2\left(\alpha - \frac{\beta+1}{2}\right)} = \frac{1}{(m^2)^{\alpha-(\beta+1)/2}}.$$

代入得最终结果:

$$\int_0^\infty \frac{x^\beta dx}{(x^2 + m^2)^\alpha} = \frac{\Gamma\left(\frac{\beta+1}{2}\right) \Gamma\left(\alpha - \frac{\beta+1}{2}\right)}{2\Gamma(\alpha) (m^2)^{\alpha-\frac{\beta+1}{2}}}.$$

Beta 函数要求 $p > 0$ 且 $q > 0$:

$$\begin{aligned} p = \frac{\beta+1}{2} > 0 &\Rightarrow \beta > -1, \\ q = \alpha - \frac{\beta+1}{2} > 0 &\Rightarrow \alpha > \frac{\beta+1}{2}. \end{aligned}$$

同时 $m > 0$ 。满足以上条件时公式成立。

考虑截断积分

$$I(\Lambda) = \int_0^\Lambda \frac{x^4}{(x^2 + m^2)^2} dx, \quad \Lambda > 0.$$

作部分分式分解:

$$\frac{x^4}{(x^2 + m^2)^2} = 1 - \frac{2m^2}{x^2 + m^2} + \frac{m^4}{(x^2 + m^2)^2}.$$

于是

$$I(\Lambda) = \int_0^\Lambda dx - 2m^2 \int_0^\Lambda \frac{dx}{x^2 + m^2} + m^4 \int_0^\Lambda \frac{dx}{(x^2 + m^2)^2}.$$

$$\begin{aligned} \int \frac{dx}{x^2 + m^2} &= \frac{1}{m} \arctan\left(\frac{x}{m}\right), \\ \int \frac{dx}{(x^2 + m^2)^2} &= \frac{x}{2m^2(x^2 + m^2)} + \frac{1}{2m^3} \arctan\left(\frac{x}{m}\right). \end{aligned}$$

代入上下限:

$$\begin{aligned} \int_0^\Lambda dx &= \Lambda, \\ \int_0^\Lambda \frac{dx}{x^2 + m^2} &= \frac{1}{m} \arctan\left(\frac{\Lambda}{m}\right), \\ \int_0^\Lambda \frac{dx}{(x^2 + m^2)^2} &= \left[\frac{\Lambda}{2m^2(\Lambda^2 + m^2)} + \frac{1}{2m^3} \arctan\left(\frac{\Lambda}{m}\right) \right]. \end{aligned}$$

因此

$$\begin{aligned} I(\Lambda) &= \Lambda - 2m^2 \cdot \frac{1}{m} \arctan\left(\frac{\Lambda}{m}\right) \\ &\quad + m^4 \left[\frac{\Lambda}{2m^2(\Lambda^2 + m^2)} + \frac{1}{2m^3} \arctan\left(\frac{\Lambda}{m}\right) \right]. \end{aligned}$$

合并含 $\arctan$ 的项:

$$\begin{aligned} I(\Lambda) &= \Lambda - 2m \arctan\left(\frac{\Lambda}{m}\right) + \frac{m^2 \Lambda}{2(\Lambda^2 + m^2)} + \frac{m}{2} \arctan\left(\frac{\Lambda}{m}\right) \\ &= \Lambda + \frac{m^2 \Lambda}{2(\Lambda^2 + m^2)} - \frac{3m}{2} \arctan\left(\frac{\Lambda}{m}\right). \end{aligned}$$

$\Lambda \rightarrow \infty$ 利用

$$\arctan\left(\frac{\Lambda}{m}\right) = \frac{\pi}{2} - \frac{m}{\Lambda} + \frac{m^3}{3\Lambda^3} + O\left(\frac{1}{\Lambda^5}\right),$$

以及

$$\frac{m^2\Lambda}{2(\Lambda^2 + m^2)} = \frac{m^2}{2\Lambda} - \frac{m^4}{2\Lambda^3} + O\left(\frac{1}{\Lambda^5}\right).$$

代入：

$$\begin{aligned} I(\Lambda) &= \Lambda + \left[\frac{m^2}{2\Lambda} - \frac{m^4}{2\Lambda^3} + \dots \right] \\ &\quad - \frac{3m}{2} \left[\frac{\pi}{2} - \frac{m}{\Lambda} + \frac{m^3}{3\Lambda^3} + \dots \right] \\ &= \Lambda + \frac{m^2}{2\Lambda} - \frac{m^4}{2\Lambda^3} - \frac{3m\pi}{4} + \frac{3m^2}{2\Lambda} - \frac{m^4}{2\Lambda^3} + O\left(\frac{1}{\Lambda^5}\right) \\ &= \Lambda - \frac{3m\pi}{4} + \frac{2m^2}{\Lambda} - \frac{m^4}{\Lambda^3} + O\left(\frac{1}{\Lambda^5}\right). \end{aligned}$$

当 $\Lambda \rightarrow \infty$ 时, $I(\Lambda) \rightarrow \infty$, 发散项为 Λ 。去掉发散项, 常数项即为有限部分 (regularized finite part):

$$\text{FP} \int_0^\infty \frac{x^4}{(x^2 + m^2)^2} dx = -\frac{3m\pi}{4}.$$

此有限值与通过 Γ 函数公式解析延拓所得结果一致

$$\int_0^\infty \frac{d^4l}{(2\pi)^4} \frac{l^2}{(l^2 + m^2)^2} \quad (52)$$

先三维后一维：

计算积分：

$$I = \int \frac{d^d l}{(2\pi)^d} \frac{l^2}{(l^2 + m^2)^2}, \quad d = 4 - 2\varepsilon,$$

其中 $l^2 = l_0^2 - \mathbf{l}^2$ 。

1: Wick 转动到欧氏空间

令 $l_0 = ip_0$, $\mathbf{l} = \mathbf{p}$, 则：

$$l^2 = -p_0^2 - \mathbf{p}^2 = -p_E^2, \quad (53)$$

$$d^d l = i d^d p_E, \quad (54)$$

其中 $p_E^2 = p_0^2 + \mathbf{p}^2$ 。

代入：

$$I = i \int \frac{d^d p_E}{(2\pi)^d} \frac{-p_E^2}{(-p_E^2 + m^2)^2} \quad (55)$$

$$= i \int \frac{d^d p_E}{(2\pi)^d} \frac{p_E^2}{(p_E^2 + m^2)^2} \equiv iI_E. \quad (56)$$

2: 分离积分

将 p_E 分为 p_0 和 $\mathbf{p}$, $p_E^2 = p_0^2 + \mathbf{p}^2$:

$$I_E = \int \frac{dp_0}{2\pi} \int \frac{d^{d-1}\mathbf{p}}{(2\pi)^{d-1}} \frac{p_0^2 + \mathbf{p}^2}{(p_0^2 + \mathbf{p}^2 + m^2)^2} \quad (57)$$

$$= \int \frac{dp_0}{2\pi} [G_1(p_0) + G_2(p_0)], \quad (58)$$

其中

$$G_1(p_0) = \int \frac{d^{d-1}\mathbf{p}}{(2\pi)^{d-1}} \frac{p_0^2}{(p_0^2 + \mathbf{p}^2 + m^2)^2},$$

$$G_2(p_0) = \int \frac{d^{d-1}\mathbf{p}}{(2\pi)^{d-1}} \frac{\mathbf{p}^2}{(p_0^2 + \mathbf{p}^2 + m^2)^2}.$$

3: 维数正规化

设 $n = d - 1 = 3 - 2\varepsilon$, $a = p_0^2 + m^2 > 0$.

$$G_1(p_0) = \frac{p_0^2}{(4\pi)^{n/2}\Gamma(n/2)} \int_0^\infty \frac{t^{n/2-1}dt}{(t+a)^2} \quad (59)$$

$$= \frac{p_0^2}{(4\pi)^{n/2}\Gamma(n/2)} \cdot a^{n/2-2} \cdot \frac{\Gamma(2-n/2)\Gamma(n/2)}{\Gamma(2)} \quad (60)$$

$$= \frac{p_0^2}{(4\pi)^{n/2}} \cdot \frac{\Gamma(2-n/2)}{\Gamma(2)} \cdot a^{n/2-2}. \quad (61)$$

$$G_2(p_0) = \int \frac{d^n\mathbf{p}}{(2\pi)^n} \frac{\mathbf{p}^2}{(p^2+a)^2} \quad (62)$$

$$= \int \frac{d^n\mathbf{p}}{(2\pi)^n} \frac{1}{p^2+a} - a \int \frac{d^n\mathbf{p}}{(2\pi)^n} \frac{1}{(p^2+a)^2} \quad (63)$$

$$= \frac{1}{(4\pi)^{n/2}} [\Gamma(1-n/2)a^{n/2-1} - a \cdot \Gamma(2-n/2)a^{n/2-2}] \quad (64)$$

$$= \frac{a^{n/2-1}}{(4\pi)^{n/2}} [\Gamma(1-n/2) - \Gamma(2-n/2)]. \quad (65)$$

利用 $\Gamma(2-n/2) = (1-n/2)\Gamma(1-n/2)$:

$$G_2(p_0) = \frac{a^{n/2-1}}{(4\pi)^{n/2}} \cdot \frac{n}{2} \cdot \Gamma(1-n/2). \quad (66)$$

$$G_1(p_0) + G_2(p_0) = \frac{1}{(4\pi)^{n/2}} \left[p_0^2 \Gamma(2-n/2) a^{n/2-2} + \frac{n}{2} \Gamma(1-n/2) a^{n/2-1} \right]. \quad (67)$$

$\Gamma(2-n/2) = (1-n/2)\Gamma(1-n/2)$, 代入:

$$= \frac{\Gamma(1-n/2)}{(4\pi)^{n/2}} a^{n/2-2} \left[p_0^2 (1-n/2) + \frac{n}{2} a \right] \quad (68)$$

$$= \frac{\Gamma(1-n/2)}{(4\pi)^{n/2}} a^{n/2-2} \left[p_0^2 - \frac{n}{2} p_0^2 + \frac{n}{2} (p_0^2 + m^2) \right] \quad (69)$$

$$= \frac{\Gamma(1-n/2)}{(4\pi)^{n/2}} a^{n/2-2} \left[p_0^2 - \frac{n}{2} p_0^2 + \frac{n}{2} p_0^2 + \frac{n}{2} m^2 \right] \quad (70)$$

$$= \frac{\Gamma(1-n/2)}{(4\pi)^{n/2}} a^{n/2-2} \left(p_0^2 + \frac{n}{2} m^2 \right). \quad (71)$$

$$I_E = \int_{-\infty}^{\infty} \frac{dp_0}{2\pi} [G_1 + G_2] \quad (72)$$

$$= \frac{\Gamma(1 - n/2)}{(4\pi)^{n/2}} \int_{-\infty}^{\infty} \frac{dp_0}{2\pi} (p_0^2 + m^2)^{n/2-2} \left(p_0^2 + \frac{n}{2} m^2 \right). \quad (73)$$

令 $\alpha = n/2 - 2 = \frac{d-5}{2}$, 则:

$$I_E = \frac{\Gamma(-\alpha - 1)}{(4\pi)^{n/2}} \int_{-\infty}^{\infty} \frac{dp_0}{2\pi} (p_0^2 + m^2)^\alpha (p_0^2 + (\alpha + 2)m^2).$$

分解:

$$(p_0^2 + m^2)^\alpha (p_0^2 + (\alpha + 2)m^2) = (p_0^2 + m^2)^{\alpha+1} + (\alpha + 1)m^2 (p_0^2 + m^2)^\alpha.$$

所以:

$$I_E = \frac{\Gamma(-\alpha - 1)}{(4\pi)^{n/2}} \left[\int_{-\infty}^{\infty} \frac{dp_0}{2\pi} (p_0^2 + m^2)^{\alpha+1} + (\alpha + 1)m^2 \int_{-\infty}^{\infty} \frac{dp_0}{2\pi} (p_0^2 + m^2)^\alpha \right].$$

4: 计算一维积分

已知:

$$\int_{-\infty}^{\infty} (p_0^2 + m^2)^\beta dp_0 = \sqrt{\pi} m^{2\beta+1} \frac{\Gamma(-\beta - 1/2)}{\Gamma(-\beta)}.$$

对于第一个积分 $\beta = \alpha + 1 = n/2 - 1 = \frac{d-3}{2}$:

$$2\beta + 1 = d - 2 = 2 - 2\varepsilon, \quad -\beta - 1/2 = 1 - \frac{d}{2} = \varepsilon, \quad -\beta = \frac{3-d}{2} = \frac{1+2\varepsilon}{2}.$$

$$\int_{-\infty}^{\infty} (p_0^2 + m^2)^{\alpha+1} dp_0 = \sqrt{\pi} m^{2-2\varepsilon} \frac{\Gamma(\varepsilon)}{\Gamma\left(\frac{1+2\varepsilon}{2}\right)}.$$

对于第二个积分 $\beta = \alpha = n/2 - 2 = \frac{d-5}{2}$:

$$2\beta + 1 = d - 4 = -2\varepsilon, \quad -\beta - 1/2 = \frac{6-d}{2} = 1 + \varepsilon, \quad -\beta = \frac{5-d}{2} = \frac{1+2\varepsilon}{2}.$$

$$\int_{-\infty}^{\infty} (p_0^2 + m^2)^\alpha dp_0 = \sqrt{\pi} m^{-2\varepsilon} \frac{\Gamma(1 + \varepsilon)}{\Gamma\left(\frac{1+2\varepsilon}{2}\right)}.$$

$$I = iI_E = \frac{im^2}{16\pi^2} \left[-\frac{2}{\varepsilon} + 2\gamma - 1 + 2\ln \frac{m^2}{4\pi\mu^2} + O(\varepsilon) \right].$$

四维维数积分:

计算积分:

$$I = \int \frac{d^d l}{(2\pi)^d} \frac{l^2}{(l^2 + m^2)^2}, \quad d = 4 - 2\varepsilon,$$

其中 $l^2 = l_0^2 - \mathbf{l}^2$ 。

Wick 转动到欧氏空间

令 $l_0 = ip_d$, 空间部分不变: $\mathbf{l} = \mathbf{p}$, 则

$$l^2 = l_0^2 - \mathbf{l}^2 = -p_d^2 - \mathbf{p}^2 = -p_E^2, \quad (74)$$

$$d^d l = i d^d p_E, \quad (75)$$

其中 $p_E^2 = p_d^2 + \mathbf{p}^2 \geq 0$ 。

代入原积分：

$$I = i \int \frac{d^d p_E}{(2\pi)^d} \frac{-p_E^2}{(-p_E^2 + m^2)^2}. \quad (76)$$

变量代换 $p_E \rightarrow ip_E$

$$I_E = \int \frac{d^d p}{(2\pi)^d} \frac{p^2}{(p^2 + m^2)^2},$$

$$I_E = \int \frac{d^d p}{(2\pi)^d} \frac{1}{p^2 + m^2} - m^2 \int \frac{d^d p}{(2\pi)^d} \frac{1}{(p^2 + m^2)^2}. \quad (77)$$

维数正规化

$$\int \frac{d^d p}{(2\pi)^d} \frac{1}{(p^2 + m^2)^A} = \frac{1}{(4\pi)^{d/2}} \frac{\Gamma(A - d/2)}{\Gamma(A)} (m^2)^{d/2 - A}.$$

分别计算：

$$I_1 = \int \frac{d^d p}{(2\pi)^d} \frac{1}{p^2 + m^2} = \frac{1}{(4\pi)^{d/2}} \Gamma\left(1 - \frac{d}{2}\right) (m^2)^{d/2 - 1}, \quad (78)$$

$$I_2 = \int \frac{d^d p}{(2\pi)^d} \frac{1}{(p^2 + m^2)^2} = \frac{1}{(4\pi)^{d/2}} \frac{\Gamma(2 - d/2)}{\Gamma(2)} (m^2)^{d/2 - 2}. \quad (79)$$

$$I_E = I_1 - m^2 I_2 \quad (80)$$

$$= \frac{1}{(4\pi)^{d/2}} \left[\Gamma\left(1 - \frac{d}{2}\right) (m^2)^{d/2 - 1} - m^2 \Gamma(2 - d/2) (m^2)^{d/2 - 2} \right] \quad (81)$$

$$= \frac{(m^2)^{d/2 - 2}}{(4\pi)^{d/2}} \left[m^2 \Gamma\left(1 - \frac{d}{2}\right) - m^2 \Gamma(2 - d/2) \right] \quad (82)$$

$$= \frac{m^2 (m^2)^{d/2 - 2}}{(4\pi)^{d/2}} \left[\Gamma\left(1 - \frac{d}{2}\right) - \Gamma(2 - d/2) \right]. \quad (83)$$

利用 Γ 函数递推关系 $\Gamma(z + 1) = z\Gamma(z)$ ：

$$\Gamma(2 - d/2) = \left(1 - \frac{d}{2}\right) \Gamma\left(1 - \frac{d}{2}\right). \quad (84)$$

代入得：

$$\Gamma\left(1 - \frac{d}{2}\right) - \Gamma(2 - d/2) = \Gamma\left(1 - \frac{d}{2}\right) - \left(1 - \frac{d}{2}\right) \Gamma\left(1 - \frac{d}{2}\right) \quad (85)$$

$$= \frac{d}{2} \Gamma\left(1 - \frac{d}{2}\right). \quad (86)$$

因此：

$$I_E = \frac{m^2 (m^2)^{d/2 - 2}}{(4\pi)^{d/2}} \cdot \frac{d}{2} \cdot \Gamma\left(1 - \frac{d}{2}\right). \quad (87)$$

$$d = 4 - 2\varepsilon$$

$$\frac{d}{2} = 2 - \varepsilon, \quad (88)$$

$$(m^2)^{d/2 - 2} = (m^2)^{\frac{4 - 2\varepsilon}{2} - 2} = (m^2)^{2 - \varepsilon - 2} = (m^2)^{-\varepsilon}, \quad (89)$$

$$(4\pi)^{d/2} = (4\pi)^{2 - \varepsilon} = (4\pi)^2 (4\pi)^{-\varepsilon} = 16\pi^2 (4\pi)^{-\varepsilon}, \quad (90)$$

$$\Gamma\left(1 - \frac{d}{2}\right) = \Gamma(1 - (2 - \varepsilon)) = \Gamma(-1 + \varepsilon). \quad (91)$$

代入：

$$I_E = \frac{m^2(m^2)^{-\varepsilon}}{16\pi^2(4\pi)^{-\varepsilon}} \cdot (2 - \varepsilon) \cdot \Gamma(-1 + \varepsilon) \quad (92)$$

$$= \frac{m^2}{16\pi^2} (4\pi)^\varepsilon m^{-2\varepsilon} (2 - \varepsilon) \Gamma(-1 + \varepsilon). \quad (93)$$

展开 ε 级数

$$\Gamma(-1 + \varepsilon) = \frac{\Gamma(1 + \varepsilon)}{(-1 + \varepsilon)\varepsilon} \quad (94)$$

$$= \frac{1}{\varepsilon(-1 + \varepsilon)} \left[1 - \gamma\varepsilon + \frac{\gamma^2 + \pi^2/6}{2} \varepsilon^2 + O(\varepsilon^3) \right]. \quad (95)$$

将 $\frac{1}{\varepsilon-1} = -1 - \varepsilon - \varepsilon^2 + O(\varepsilon^3)$ 代入：

$$\Gamma(-1 + \varepsilon) = -\frac{1}{\varepsilon} (1 + \varepsilon + \varepsilon^2) \left[1 - \gamma\varepsilon + \frac{\gamma^2 + \pi^2/6}{2} \varepsilon^2 \right] + O(\varepsilon^2) \quad (96)$$

$$= -\frac{1}{\varepsilon} \left[1 + (1 - \gamma)\varepsilon + \left(1 - \gamma + \frac{\gamma^2 + \pi^2/6}{2} \right) \varepsilon^2 \right] + O(\varepsilon^2) \quad (97)$$

$$= -\frac{1}{\varepsilon} + (\gamma - 1) + O(\varepsilon). \quad (98)$$

$$(4\pi)^\varepsilon = e^{\varepsilon \ln(4\pi)} = 1 + \varepsilon \ln(4\pi) + \frac{\varepsilon^2}{2} [\ln(4\pi)]^2 + O(\varepsilon^3), \quad (99)$$

$$m^{-2\varepsilon} = e^{-2\varepsilon \ln m} = 1 - 2\varepsilon \ln m + 2\varepsilon^2 (\ln m)^2 + O(\varepsilon^3), \quad (100)$$

整理结果：

$$I_E = \frac{m^2}{16\pi^2} [1 + \varepsilon \ln(4\pi)] [1 - 2\varepsilon \ln m] (2 - \varepsilon) \left[-\frac{1}{\varepsilon} + (\gamma - 1) + O(\varepsilon) \right]. \quad (101)$$

$$I_E = \frac{m^2}{16\pi^2} \left\{ \left[-\frac{2}{\varepsilon} + 2\gamma - 1 \right] + \varepsilon \ln \frac{4\pi}{m^2} \cdot \left(-\frac{2}{\varepsilon} \right) \right\} + O(\varepsilon) \quad (102)$$

$$= \frac{m^2}{16\pi^2} \left[-\frac{2}{\varepsilon} + 2\gamma - 1 - 2 \ln \frac{4\pi}{m^2} \right] + O(\varepsilon). \quad (103)$$

回到闵氏积分

$$I = iI_E \quad (104)$$

$$= \frac{im^2}{16\pi^2} \left[-\frac{2}{\varepsilon} + 2\gamma - 1 - 2 \ln \frac{4\pi}{m^2} \right] + O(\varepsilon). \quad (105)$$

引入重正化标度 μ (令 $m^2 \rightarrow m^2/\mu^2$):

$$-2 \ln \frac{4\pi}{m^2} \rightarrow -2 \ln \frac{4\pi\mu^2}{m^2} = 2 \ln \frac{m^2}{4\pi\mu^2}.$$

于是：

$$I = \frac{im^2}{16\pi^2} \left[-\frac{2}{\varepsilon} + 2\gamma - 1 + 2 \ln \frac{m^2}{4\pi\mu^2} \right] + O(\varepsilon). \quad (106)$$

考虑质量差后的圈积分公式：

$$V^{Triangle}(q, m_1, m_2, \Delta) = \int \frac{d^3 l}{(2\pi)^3} \frac{l^2 - q^2}{(\omega_+ + 2\Delta)(\omega_- + 2\Delta)(\omega_+ + \omega_-)} \quad (107)$$

$$\mu^{4-d} \int \frac{d^{d-1} l}{(2\pi)^{d-1}} \frac{l^2 - q^2}{(\omega_+ + 2\Delta)(\omega_- + 2\Delta)(\omega_+ + \omega_-)} \quad (108)$$

$$= \frac{4\Delta^2}{\pi} \int_{-\infty}^{\infty} \frac{d\lambda}{4\Delta^2 + \lambda^2} \mu^{4-d} \int \frac{d^{d-1} l}{(2\pi)^{d-1}} \frac{l^2 - q^2}{(\omega_+^2 + \lambda^2)(\omega_-^2 + \lambda^2)} \quad (109)$$

$$= \frac{4\Delta^2}{\pi} \int_{-\infty}^{\infty} \frac{d\lambda}{4\Delta^2 + \lambda^2} \mu^{4-d} \int \frac{d^{d-1} l}{(2\pi)^{d-1}} \frac{l^2 - q^2}{((l+q)^2 + 4m_1^2 + \lambda^2)((l-q)^2 + 4m_2^2 + \lambda^2)} \quad (110)$$

首先处理 $d-1$ 维的积分，利用费曼参数化：

$$\mu^{4-d} \int \frac{d^{d-1} l}{(2\pi)^{d-1}} \frac{l^2 - q^2}{((l+q)^2 + 4m_1^2 + \lambda^2)((l-q)^2 + 4m_2^2 + \lambda^2)} \quad (111)$$

$$= \int_0^1 dx \mu^{4-d} \int \frac{d^{d-1} l}{(2\pi)^{d-1}} \frac{l^2 - q^2}{[x((l+q)^2 + 4m_1^2 + \lambda^2) + (1-x)((l-q)^2 + 4m_2^2 + \lambda^2)]^2} \quad (112)$$

$$= \int_0^1 dx \mu^{4-d} \int \frac{d^{d-1} l}{(2\pi)^{d-1}} \frac{l^2 - q^2}{[(l + (2x-1)q)^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + \lambda^2]^2} \quad (113)$$

进行积分变量平移变换 $l \rightarrow l - (2x-1)q$ ：

$$= \int_0^1 dx \mu^{4-d} \int \frac{d^{d-1} l}{(2\pi)^{d-1}} \frac{l^2 - 2(2x-1)l \cdot q - 4x(1-x)q^2}{[l^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + \lambda^2]^2} \quad (114)$$

$$= \int_0^1 dx \mu^{4-d} \int \frac{d^{d-1} l}{(2\pi)^{d-1}} \frac{l^2 - 4x(1-x)q^2}{[l^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + \lambda^2]^2} \quad (115)$$

被积函数中， l 的奇次幂项为奇函数，可以去掉。对于第一项：

$$\int_0^1 dx \mu^{4-d} \int \frac{d^{d-1} l}{(2\pi)^{d-1}} \frac{l^2}{[l^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + \lambda^2]^2} \quad (116)$$

$$= \int_0^1 dx \frac{2\mu^{4-d}}{(4\pi)^{\frac{d-1}{2}} \Gamma(\frac{d-1}{2})} \int_0^\infty \frac{l^d dl}{[l^2 + M^2]^2} \quad (117)$$

$$= \int_0^1 dx \frac{2\mu^{4-d}}{(4\pi)^{\frac{d-1}{2}} \Gamma(\frac{d-1}{2})} \frac{\Gamma(\frac{d+1}{2}) \Gamma(\frac{3-d}{2})}{2(M^2)^{\frac{3-d}{2}}} \quad (118)$$

$$= \int_0^1 dx \frac{\mu^{4-d}(d-1)}{2(4\pi)^{\frac{d-1}{2}}} \frac{\Gamma(\frac{3-d}{2})}{(M^2)^{\frac{3-d}{2}}}$$

其中 $M^2 = 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + \lambda^2$ 。

对于第二项：

$$- 4q^2 \int_0^1 dx x(1-x) \mu^{4-d} \int \frac{d^{d-1} l}{(2\pi)^{d-1}} \frac{1}{[l^2 + M^2]^2} \quad (119)$$

$$= -4q^2 \int_0^1 dx \frac{x(1-x)\mu^{4-d}}{(4\pi)^{\frac{d-1}{2}}} \frac{\Gamma(\frac{5-d}{2})}{(M^2)^{\frac{5-d}{2}}} \quad (120)$$

带回到 (115) 式，对于第一项：

$$-\frac{2\Delta^2}{\pi} \int_0^1 dx \frac{\mu^{4-d}(d-1)\Gamma(\frac{3-d}{2})}{(4\pi)^{\frac{d-1}{2}}} \int_{-\infty}^{\infty} \frac{d\lambda}{(4\Delta^2 + \lambda^2)[M^2 + \lambda^2]^{\frac{3-d}{2}}} \quad (121)$$

$$= -\frac{2\Delta^2}{\pi} \int_0^1 dx \frac{\mu^{4-d}(d-1)\Gamma(\frac{3-d}{2})}{(4\pi)^{\frac{d-1}{2}}} \frac{1}{4} (M^2)^{-(\frac{3-d}{2})} \sqrt{\pi} \left[\frac{\Gamma(\frac{2-d}{2}) {}_2F_1(\frac{1}{2}, 1; \frac{3}{2} - (\frac{3-d}{2}); \frac{M^2}{4\Delta^2})}{\sqrt{\frac{1}{M^2} \Delta^2 \Gamma((\frac{3-d}{2}))}} \right]$$

$$+ 2\left(\frac{1}{M^2}\right)^{-(\frac{3-d}{2})} \sqrt{\pi} \left(4 - \frac{M^2}{\Delta^2}\right)^{-(\frac{3-d}{2})} \left(\frac{1}{\Delta^2}\right)^{\frac{1}{2} + (\frac{3-d}{2})} \sec\left(\pi\left(\frac{3-d}{2}\right)\right) \quad (122)$$

$$= -\frac{\Delta^2}{4\pi^2} \int_0^1 dx \left[2(3R + 2 + 3\ln(\frac{4\Delta^2}{\mu^2})) + F_1(z(x)) \right] + O(\epsilon^1) \quad (123)$$

其中超几何函数：

$${}_2F_1(a, b; c; z) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 dt \frac{t^{b-1}(1-t)^{c-b-1}}{(1-tz)^a}$$

其中 $z = M^2/(4\Delta^2)$, 且

$$F_1(z) = 3z {}_2F_1\left(\frac{1}{2}, 1; 2; z\right) + 6 \left[\ln(z) + \sqrt{1-z} \left(\ln\left(\frac{4}{z} - 4\right) - 1 \right) \right]$$

对于第二项：

$$\frac{16\Delta^2 q^2}{\pi} \int_0^1 dx \frac{x(1-x)\mu^{4-d}\Gamma(\frac{5-d}{2})}{(4\pi)^{\frac{d-1}{2}}} \int_{-\infty}^{\infty} \frac{d\lambda}{(4\Delta^2 + \lambda^2)[M^2 + \lambda^2]^{\frac{5-d}{2}}} \quad (124)$$

$$= -\frac{q^2}{2\pi^2} \int_0^1 dx x(1-x) F_2(z) + O(\epsilon^1) \quad (125)$$

其中

$$F_2(z) = {}_2F_1\left(\frac{1}{2}, 1; 1; z\right) - \frac{1}{\sqrt{1-z}} \ln\left(\frac{4}{z} - 4\right)$$

三角图中圈积分维数正规化的结果可以写为：

$$-\frac{\Delta^2}{4\pi^2} \left[2 \left(3R + 2 + 3\ln\left(\frac{4\Delta^2}{\mu^2}\right) \right) + \int_0^1 dx F_1(z(x)) \right] - \frac{q^2}{2\pi^2} \int_0^1 dx x(1-x) F_2(z(x)) \quad (126)$$

平行箱图：

$$V^{\text{PlanarBox}}(q, m_1, m_2, \Delta) = -\frac{1}{8} \int \frac{d^3 l}{(2\pi)^3} \frac{(l^2 - q^2)^2}{\omega_+ \omega_- (\omega_+ + 2\Delta)^2 (\omega_- + 2\Delta)^2 (\omega_+ + \omega_-)} \quad (127)$$

交叉箱图：

$$V^{\text{CrossBox}}(q, m_1, m_2, \Delta) = -\frac{1}{8} \int \frac{d^3 l}{(2\pi)^3} \frac{((\omega_+ + 2\Delta)^2 + \omega_+ \omega_- - 4\Delta^2 + (\omega_- + 2\Delta)^2)(l^2 - q^2)^2}{\omega_+ \omega_- (\omega_+ + 2\Delta)^2 (\omega_- + 2\Delta)^2 (\omega_+ + \omega_-)} \quad (128)$$

计算交叉箱图的圈积分结果：

$$\begin{aligned}
& \frac{(\omega_+ + 2\Delta)^2 + \omega_+ \omega_- - 4\Delta^2 + (\omega_- + 2\Delta)^2}{\omega_+ \omega_- (\omega_+ + 2\Delta)^2 (\omega_- + 2\Delta)^2 (\omega_+ + \omega_-)} \\
&= -\frac{1}{2} \left(\frac{\partial}{\partial m_1^2} + \frac{\partial}{\partial m_2^2} \right) \frac{1}{(\omega_+ + 2\Delta)(\omega_- + 2\Delta)(\omega_+ + \omega_-)} \\
&= -\frac{1}{2\pi} \left(\frac{\partial}{\partial m_1^2} + \frac{\partial}{\partial m_2^2} \right) \int_{-\infty}^{\infty} \frac{d\lambda}{(\omega_+^2 + \lambda^2)(\omega_-^2 + \lambda^2)(4\Delta^2 + \lambda^2)} \\
&= -\frac{1}{2\pi} \left(\frac{\partial}{\partial m_1^2} + \frac{\partial}{\partial m_2^2} \right) \int_{-\infty}^{\infty} \frac{d\lambda}{(\omega_+^2 + \lambda^2)(\omega_-^2 + \lambda^2)} \\
&\quad + \frac{2\Delta^2}{\pi} \left(\frac{\partial}{\partial m_1^2} + \frac{\partial}{\partial m_2^2} \right) \int_{-\infty}^{\infty} \frac{d\lambda}{(\omega_+^2 + \lambda^2)(\omega_-^2 + \lambda^2)(4\Delta^2 + \lambda^2)} \tag{129}
\end{aligned}$$

第二项是质量差带来的修正。对于交叉箱图

$$\mu^{4-d} \int \frac{d^{d-1}l}{(2\pi)^{d-1}} \frac{((\omega_+ + 2\Delta)^2 + \omega_+ \omega_- - 4\Delta^2 + (\omega_- + 2\Delta)^2)(l^2 - q^2)^2}{\omega_+ \omega_- (\omega_+ + 2\Delta)^2 (\omega_- + 2\Delta)^2 (\omega_+ + \omega_-)} \tag{130}$$

根据 (129) 式分解成两项为

$$\begin{aligned}
& -\frac{1}{2\pi} \left(\frac{\partial}{\partial m_1^2} + \frac{\partial}{\partial m_2^2} \right) \mu^{4-d} \int_{-\infty}^{\infty} d\lambda \int \frac{d^{d-1}l}{(2\pi)^{d-1}} \frac{(l^2 - q^2)^2}{(\omega_+^2 + \lambda^2)(\omega_-^2 + \lambda^2)} \\
& + \frac{2\Delta^2}{\pi} \left(\frac{\partial}{\partial m_1^2} + \frac{\partial}{\partial m_2^2} \right) \int_{-\infty}^{\infty} \frac{d\lambda}{4\Delta^2 + \lambda^2} \mu^{4-d} \int \frac{d^{d-1}l}{(2\pi)^{d-1}} \frac{(l^2 - q^2)^2}{(\omega_+^2 + \lambda^2)(\omega_-^2 + \lambda^2)} \tag{131}
\end{aligned}$$

第一项已经处理过，讨论第二项的计算，首先计算动量积分

$$\begin{aligned}
& \mu^{4-d} \int \frac{d^{d-1}l}{(2\pi)^{d-1}} \frac{(l^2 - q^2)^2}{(\omega_+^2 + \lambda^2)(\omega_-^2 + \lambda^2)} \\
&= \mu^{4-d} \int \frac{d^{d-1}l}{(2\pi)^{d-1}} \frac{(l^2 - q^2)^2}{((l+q)^2 + 4m_1^2 + \lambda^2)((l-q)^2 + 4m_2^2 + \lambda^2)} \\
&= \int_0^1 dx \mu^{4-d} \int \frac{d^{d-1}l}{(2\pi)^{d-1}} \frac{(l^2 - q^2)^2}{[l^2 + (2x-1)q^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + \lambda^2]^2} \\
&= \int_0^1 dx \mu^{4-d} \int \frac{d^{d-1}l}{(2\pi)^{d-1}} \frac{[l^2 - 2(2x-1)l \cdot q - 4x(1-x)q^2]^2}{[l^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + \lambda^2]^2} \tag{132}
\end{aligned}$$

去掉 l 奇次幂项

$$= \int_0^1 dx \mu^{4-d} \int \frac{d^{d-1}l}{(2\pi)^{d-1}} \frac{l^4 - 8x(1-x)l^2 q^2 + 4(2x-1)^2(l \cdot q)^2 + 16x^2(1-x)^2 q^4}{[l^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + \lambda^2]^2} \tag{133}$$

式中包含四项，对于第一项

$$\begin{aligned}
& \int_0^1 dx \mu^{4-d} \int \frac{d^{d-1}l}{(2\pi)^{d-1}} \frac{l^4}{[l^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + \lambda^2]^2} \\
&= \int_0^1 dx \frac{\mu^{4-d}(d-1)(d+1)\Gamma\left(\frac{1-d}{2}\right)}{4(4\pi)^{\frac{d-1}{2}}} \frac{1}{[l^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + \lambda^2]^{\frac{1-d}{2}}} \tag{134}
\end{aligned}$$

带回 (131) 式

$$\begin{aligned}
& \frac{2\Delta^2}{\pi} \left(\frac{\partial}{\partial m_1^2} + \frac{\partial}{\partial m_2^2} \right) \int_0^1 dx \frac{\mu^{4-d}(d-1)(d+1)\Gamma\left(\frac{1-d}{2}\right)}{4(4\pi)^{\frac{d-1}{2}}} \\
& \times \int_{-\infty}^{\infty} \frac{d\lambda}{(4\Delta^2 + \lambda^2)[l^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + \lambda^2]^{\frac{1-d}{2}}} \\
& = \frac{\Delta^2}{8} \left(\frac{\partial}{\partial m_1^2} + \frac{\partial}{\partial m_2^2} \right) \int_0^1 dx \frac{\mu^{4-d}(d-1)(d+1)\Gamma\left(\frac{1-d}{2}\right)}{(4\pi)^{\frac{d-1}{2}}} \csc\left(\frac{\pi d}{2}\right) \\
& \times \left[2\Delta^{d-2} \left(4 - \frac{M^2}{\Delta^2}\right)^{\frac{d-1}{2}} - \frac{\sqrt{\pi}(M^2)^{\frac{d}{2}} {}_2F_1\left(\frac{1}{2}, 1; \frac{d}{2} + 1; \frac{M^2}{4\Delta^2}\right)}{\Delta^2 \Gamma\left(\frac{1-d}{2}\right) \Gamma\left(\frac{d}{2} + 1\right)} \right] \\
& = \frac{\Delta^2}{8} \left(\frac{\partial}{\partial m_1^2} + \frac{\partial}{\partial m_2^2} \right) \int_0^1 dx \frac{\mu^\epsilon(3-\epsilon)(5-\epsilon)\Gamma\left(\frac{\epsilon-3}{2}\right)}{(4\pi)^{\frac{3-\epsilon}{2}}} \csc\left(\frac{\pi(4-\epsilon)}{2}\right) \\
& \times \left[2\Delta^{2-\epsilon} \left(4 - \frac{M^2}{\Delta^2}\right)^{\frac{3-\epsilon}{2}} - \frac{\sqrt{\pi}(M^2)^{\frac{4-\epsilon}{2}} {}_2F_1\left(\frac{1}{2}, 1; 3 - \frac{\epsilon}{2}; \frac{M^2}{4\Delta^2}\right)}{\Delta^2 \Gamma\left(\frac{\epsilon-3}{2}\right) \Gamma\left(3 - \frac{\epsilon}{2}\right)} \right] \\
& = -\frac{\Delta^4}{24\pi^2} \left(\frac{\partial}{\partial m_1^2} + \frac{\partial}{\partial m_2^2} \right) \int_0^1 dx \left\{ 60 \left(3\frac{M^2}{4\Delta^2} - 2 \right) \left[R + \ln\left(\frac{4\Delta^2}{\mu^2}\right) \right] \right. \\
& \quad \left. - 68 + 102\frac{M^2}{4\Delta^2} + F_3(z(x)) \right\} + O(\epsilon^1)
\end{aligned} \tag{135}$$

其中

$$F_3(z) = 45z^2 {}_2F_1\left(\frac{1}{2}, 1; 3; z\right) + 60(3z-2)\ln(z) + 140(1-z)^{3/2} - 120(1-z)^{3/2}\ln\left(\frac{4}{z} - 4\right) \tag{136}$$

$$\left(\frac{\partial}{\partial m_1^2} + \frac{\partial}{\partial m_2^2} \right) M^2 = x + (1-x) = 1 \tag{139}$$

$$\left(\frac{\partial}{\partial m_1^2} + \frac{\partial}{\partial m_2^2} \right) F_3(z(x)) = F_3'(z) \frac{\partial z}{\partial m_1^2} + F_3'(z) \frac{\partial z}{\partial m_2^2} = \frac{F_3'(z)}{4\Delta^2} \tag{140}$$

去掉不依赖于 M^2 项, 最终这项的贡献为

$$-\frac{\Delta^2}{96\pi^2} \int_0^1 dx \left\{ 180 \left[R + \ln\left(\frac{4\Delta^2}{\mu^2}\right) \right] + 102 + F_3'(z) \right\} \tag{137}$$

对于第二项

$$\begin{aligned}
& -8q^2 \int_0^1 dx x(1-x) \mu^{4-d} \int \frac{d^{d-1}l}{(2\pi)^{d-1}} \frac{l^2}{[l^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + \lambda^2]^2} \\
& = -8q^2 \int_0^1 dx x(1-x) \frac{\mu^{4-d}(d-1)\Gamma\left(\frac{3-d}{2}\right)}{2(4\pi)^{\frac{d-1}{2}}} \frac{1}{[l^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + \lambda^2]^{\frac{3-d}{2}}}
\end{aligned} \tag{138}$$

带回 (131) 式

$$\begin{aligned}
& -\frac{8q^2\Delta^2}{\pi} \left(\frac{\partial}{\partial m_1^2} + \frac{\partial}{\partial m_2^2} \right) \int_0^1 dx x(1-x) \frac{\mu^{4-d}(d-1)\Gamma\left(\frac{3-d}{2}\right)}{(4\pi)^{\frac{d-1}{2}}} \\
& \times \int_{-\infty}^{\infty} \frac{d\lambda}{(4\Delta^2 + \lambda^2)[l^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + \lambda^2]^{\frac{3-d}{2}}} \\
& = -\frac{q^2\Delta^2}{\pi^2} \left(\frac{\partial}{\partial m_1^2} + \frac{\partial}{\partial m_2^2} \right) \int_0^1 dx x(1-x) \left[2 \left(3R + 2 + 3\ln\left(\frac{4\Delta^2}{\mu^2}\right) \right) + F_1(z(x)) \right] + O(\epsilon^1) \\
& = -\frac{q^2}{4\pi^2} \int_0^1 dx x(1-x) F_1'(z)
\end{aligned} \tag{139}$$

第三项

$$\begin{aligned}
& 4 \int_0^1 dx (2x-1)^2 \mu^{4-d} \int \frac{d^{d-1}l}{(2\pi)^{d-1}} \frac{(l \cdot q)^2}{[l^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + \lambda^2]^2} \\
& = 4q^2 \int_0^1 dx (2x-1)^2 \frac{\mu^{4-d}}{d-1} \int \frac{d^{d-1}l}{(2\pi)^{d-1}} \frac{l^2}{[l^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + \lambda^2]^2} \\
& = 4q^2 \int_0^1 dx (2x-1)^2 \frac{\mu^{4-d} \Gamma\left(\frac{3-d}{2}\right)}{2(4\pi)^{\frac{d-1}{2}}} \frac{1}{[l^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + \lambda^2]^{\frac{3-d}{2}}}
\end{aligned} \tag{140}$$

其中利用了

$$\int \frac{d^{d-1}l}{(2\pi)^{d-1}} \frac{l_i l_j}{[l^2 + M^2]^2} = \frac{\delta_{ij}}{d-1} \int \frac{d^{d-1}l}{(2\pi)^{d-1}} \frac{l^2}{[l^2 + M^2]^2} \tag{141}$$

帶回到 (131) 式

$$\begin{aligned}
& \frac{4q^2 \Delta^2}{\pi} \left(\frac{\partial}{\partial m_1^2} + \frac{\partial}{\partial m_2^2} \right) \int_0^1 dx (2x-1)^2 \frac{\mu^{4-d} \Gamma\left(\frac{3-d}{2}\right)}{(4\pi)^{\frac{d-1}{2}}} \\
& \times \int_{-\infty}^{\infty} \frac{d\lambda}{(4\Delta^2 + \lambda^2)[l^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + \lambda^2]^{\frac{3-d}{2}}} \\
& = \frac{q^2}{2\pi^2} \left(\frac{\partial}{\partial m_1^2} + \frac{\partial}{\partial m_2^2} \right) \int_0^1 dx (2x-1)^2 \left[2 \left(R + \ln \left(\frac{4\Delta^2}{\mu^2} \right) \right) + \frac{1}{3} F_1(z(x)) \right] + O(\epsilon^1) \\
& = \frac{q^2}{24\pi^2} \int_0^1 dx (2x-1)^2 F_1'(z)
\end{aligned} \tag{142}$$

对于第四项

$$\begin{aligned}
& 16q^4 \int_0^1 dx x^2 (1-x)^2 \mu^{4-d} \int \frac{d^{d-1}l}{(2\pi)^{d-1}} \frac{1}{[l^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + \lambda^2]^2} \\
& = 16q^4 \int_0^1 dx x^2 (1-x)^2 \frac{\mu^{4-d} \Gamma\left(\frac{5-d}{2}\right)}{(4\pi)^{\frac{d-1}{2}}} \frac{1}{[l^2 + 4x(1-x)q^2 + 4m_2^2 + \lambda^2]^{\frac{5-d}{2}}}
\end{aligned} \tag{143}$$

帶回到 (131) 式

$$\begin{aligned}
& \frac{32q^4 \Delta^2}{\pi} \left(\frac{\partial}{\partial m_1^2} + \frac{\partial}{\partial m_2^2} \right) \int_0^1 dx x^2 (1-x)^2 \frac{\mu^{4-d} \Gamma\left(\frac{5-d}{2}\right)}{(4\pi)^{\frac{d-1}{2}}} \\
& \times \int_{-\infty}^{\infty} \frac{d\lambda}{(4\Delta^2 + \lambda^2)[l^2 + 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + \lambda^2]^{\frac{5-d}{2}}} \\
& = -\frac{q^4}{2\pi^2} \left(\frac{\partial}{\partial m_1^2} + \frac{\partial}{\partial m_2^2} \right) \int_0^1 dx x^2 (1-x)^2 F_2(z) + O(\epsilon^1) \\
& = -\frac{q^4}{4\Delta^2 \pi^2} \int_0^1 dx x^2 (1-x)^2 F_2'(z)
\end{aligned} \tag{144}$$

平行箱图的圈积分定义为

$$V^{\text{PlanarBox}}(q, m_1, m_2; \Delta) = -\frac{1}{8} \int \frac{d^3l}{(2\pi)^3} \frac{(l^2 - q^2)^2}{\omega_+ \omega_- (\omega_+ + 2\Delta)(\omega_- + 2\Delta)(\omega_+ + \omega_-)}, \tag{145}$$

其中 $\omega_{\pm} = \sqrt{(l \pm q)^2 + 4m^2}$, $q = |\mathbf{p}' - \mathbf{p}|$ 为动量转移, $\Delta = m_{D^*} - m_D$ 为质量差。为进行维数正规化, 将积分维数推广至 $d-1$ 维, $d \rightarrow 4$:

$$V^{\text{PlanarBox}} = -\frac{1}{8} \mu^{4-d} \int \frac{d^{d-1}l}{(2\pi)^{d-1}} \frac{(l^2 - q^2)^2}{\omega_+ \omega_- (\omega_+ + 2\Delta)(\omega_- + 2\Delta)(\omega_+ + \omega_-)}. \tag{146}$$

利用恒等式

$$\frac{1}{(\omega_+ + 2\Delta)(\omega_- + 2\Delta)(\omega_+ + \omega_-)} = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{d\lambda}{(\omega_+^2 + \lambda^2)(\omega_-^2 + \lambda^2)} - \frac{4\Delta^2}{\pi} \int_{-\infty}^{\infty} \frac{d\lambda}{(\omega_+^2 + \lambda^2)(\omega_-^2 + \lambda^2)(4\Delta^2 + \lambda^2)}, \quad (147)$$

$V^{(\Delta)}$:

$$V^{(\Delta)} = \frac{\Delta^2}{2\pi} \int_{-\infty}^{\infty} \frac{d\lambda}{4\Delta^2 + \lambda^2} \cdot \mu^{4-d} \int \frac{d^{d-1}l}{(2\pi)^{d-1}} \frac{(l^2 - q^2)^2}{\omega_+ \omega_- (\omega_+^2 + \lambda^2)(\omega_-^2 + \lambda^2)}. \quad (148)$$

对动量积分部分, 令

$$\mathcal{I} \equiv \mu^{4-d} \int \frac{d^{d-1}l}{(2\pi)^{d-1}} \frac{(l^2 - q^2)^2}{(\omega_+^2 + \lambda^2)(\omega_-^2 + \lambda^2)}, \quad (149)$$

利用 Feynman 参数化并作平移变换 $l \rightarrow l - (2x - 1)q$, 去掉 l 的奇次幂项后, 分子展开为四项:

$$\mathcal{I} = \int_0^1 dx \mu^{4-d} \int \frac{d^{d-1}l}{(2\pi)^{d-1}} \frac{l^4 - 8x(1-x)l^2q^2 + 4(2x-1)^2(l \cdot q)^2 + 16x^2(1-x)^2q^4}{[l^2 + \mathcal{M}^2]^2}, \quad (150)$$

其中

$$\mathcal{M}^2 = 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2 + \lambda^2. \quad (151)$$

第一项: l^4

$$\begin{aligned} \mathcal{I}_1 &= \int_0^1 dx \frac{2\mu^{4-d}}{(4\pi)^{\frac{d-1}{2}} \Gamma(\frac{d-1}{2})} \int_0^\infty \frac{l^{d+2} dl}{[l^2 + \mathcal{M}^2]^2} \\ &= \int_0^1 dx \frac{\mu^{4-d}(d-1)(d+1)\Gamma(\frac{1-d}{2})}{4(4\pi)^{\frac{d-1}{2}}} \frac{1}{[\mathcal{M}^2]^{\frac{1-d}{2}}}. \end{aligned} \quad (152)$$

第二项: $-8x(1-x)q^2 l^2$

$$\mathcal{I}_2 = -8q^2 \int_0^1 dx x(1-x) \frac{\mu^{4-d}(d-1)\Gamma(\frac{3-d}{2})}{2(4\pi)^{\frac{d-1}{2}}} \frac{1}{[\mathcal{M}^2]^{\frac{3-d}{2}}}. \quad (153)$$

第三项: $4(2x-1)^2(l \cdot q)^2$

$$\int \frac{d^{d-1}l}{(2\pi)^{d-1}} \frac{l_i l_j}{[l^2 + \mathcal{M}^2]^2} = \frac{\delta_{ij}}{d-1} \int \frac{d^{d-1}l}{(2\pi)^{d-1}} \frac{l^2}{[l^2 + \mathcal{M}^2]^2}, \quad (154)$$

故 $(l \cdot q)^2 \rightarrow l^2 q^2 / (d-1)$, 因此

$$\mathcal{I}_3 = 4q^2 \int_0^1 dx (2x-1)^2 \frac{\mu^{4-d}\Gamma(\frac{3-d}{2})}{2(4\pi)^{\frac{d-1}{2}}} \frac{1}{[\mathcal{M}^2]^{\frac{3-d}{2}}}. \quad (155)$$

第四项: $16x^2(1-x)^2q^4$

$$\mathcal{I}_4 = 16q^4 \int_0^1 dx x^2(1-x)^2 \frac{\mu^{4-d}\Gamma(\frac{5-d}{2})}{(4\pi)^{\frac{d-1}{2}}} \frac{1}{[\mathcal{M}^2]^{\frac{5-d}{2}}}. \quad (156)$$

对 λ 的积分利用公式

$$\int_{-\infty}^{\infty} \frac{d\lambda}{(4\Delta^2 + \lambda^2)(M^2 + \lambda^2)^\alpha} = \frac{\pi}{4} \sec(\pi\alpha) \left[2\Delta^{-2\alpha-1} \left(4 - \frac{M^2}{\Delta^2} \right)^{-\alpha} - \frac{\sqrt{\pi}(M^2)^{\frac{1}{2}-\alpha} {}_2F_1\left(\frac{1}{2}, 1; \frac{3}{2} - \alpha; \frac{M^2}{4\Delta^2}\right)}{\Delta^2 \Gamma(\frac{3}{2} - \alpha) \Gamma(\alpha)} \right], \quad (157)$$

其中 $M^2 = 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2$ 。

第一项的 λ 积分 将 $\mathcal{I}_1$ 代入 $V^{(\Delta)}$, 取 $\alpha = (1-d)/2$, 令 $d = 4 - \epsilon$, 展开至 $O(\epsilon^0)$, 利用

$$\left(\frac{\partial}{\partial m_1^2} + \frac{\partial}{\partial m_2^2}\right) M^2 = 1, \quad \left(\frac{\partial}{\partial m_1^2} + \frac{\partial}{\partial m_2^2}\right) F_3(z) = \frac{F_3'(z)}{4\Delta^2}, \quad (158)$$

最终第一项的贡献为

$$\boxed{-\frac{\Delta^2}{96\pi^2} \int_0^1 dx \left[180 \left(R + \ln \frac{4\Delta^2}{\mu^2} \right) + 102 + F_3'(z) \right]}, \quad (159)$$

其中 $R = 2/\epsilon - \gamma_E + \ln(4\pi)$, $z = M^2/(4\Delta^2)$, $F_3(z)$ 定义为

$$F_3(z) = 45z^2 {}_2F_1^{(0,0,1,0)}\left(\frac{1}{2}, 1; 3; z\right) + 60(3z-2) \ln z + 140(1-z)^{3/2} - 120(1-z)^{3/2} \ln\left(\frac{4}{z} - 4\right). \quad (160)$$

第二项的 λ 积分 将 $\mathcal{I}_2$ 代入 $V^{(\Delta)}$, 取 $\alpha = (3-d)/2$, 利用 $(\partial/\partial m_1^2 + \partial/\partial m_2^2)F_1(z) = F_1'(z)/(4\Delta^2)$,

$$\boxed{-\frac{q^2}{4\pi^2} \int_0^1 dx x(1-x) F_1'(z)}, \quad (161)$$

其中

$$F_1(z) = 3z {}_2F_1^{(0,0,1,0)}\left(\frac{1}{2}, 1; 2; z\right) + 6 \ln z + \sqrt{1-z} \left(\ln \frac{4}{z} - 4 \right) - 1. \quad (162)$$

第三项的 λ 积分 将 $\mathcal{I}_3$ 代入 $V^{(\Delta)}$,

$$\boxed{+\frac{q^2}{24\pi^2} \int_0^1 dx (2x-1)^2 F_1'(z)}. \quad (163)$$

第四项的 λ 积分 将 $\mathcal{I}_4$ 代入 $V^{(\Delta)}$, 取 $\alpha = (5-d)/2$, 利用 $(\partial/\partial m_1^2 + \partial/\partial m_2^2)F_2(z) = F_2'(z)/(4\Delta^2)$,

$$\boxed{-\frac{q^4}{4\Delta^2\pi^2} \int_0^1 dx x^2(1-x)^2 F_2'(z)}, \quad (164)$$

其中

$$F_2(z) = {}_2F_1^{(0,0,1,0)}\left(\frac{1}{2}, 1; 1; z\right) - \frac{1}{\sqrt{1-z}} \ln\left(\frac{4}{z} - 4\right). \quad (165)$$

第二三项的 q^2 系数可以合并:

$$x(1-x) - \frac{(2x-1)^2}{6} = \frac{6x(1-x) - (2x-1)^2}{6} = \frac{-10x^2 + 10x - 1}{6}. \quad (166)$$

最终平行箱图的圈积分在维数正规化框架下的完整结果为

$$\begin{aligned} V^{\text{PlanarBox}}(q, m_1, m_2; \Delta) &= \frac{\epsilon^i \cdot \epsilon^{\dagger j} g^{ij}}{3} [V^{(0)}(q, m_1, m_2) \\ &\quad + \frac{\Delta^2}{768\pi^2} \int_0^1 dx \left[180 \left(R + \ln \frac{4\Delta^2}{\mu^2} \right) + 102 + F_3'(z) \right] \\ &\quad + \frac{q^2}{192\pi^2} \int_0^1 dx (-10x^2 + 10x - 1) F_1'(z) \\ &\quad - \frac{q^4}{32\Delta^2\pi^2} \int_0^1 dx x^2(1-x)^2 F_2'(z)], \end{aligned} \quad (167)$$

其中

$$z = \frac{M^2}{4\Delta^2}, \quad M^2 = 4x(1-x)q^2 + 4(m_1^2 - m_2^2)x + 4m_2^2, \quad (168)$$

超几何函数的偏导数定义为 ${}_2F_1^{(0,0,1,0)}(a, b; c; z) = \partial_c {}_2F_1(a, b; c; z)$